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Flying Operations

F-15 – OPERATIONS PROCEDURES

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This publication establishes effective and safe operations of the F-15 and implements Air Force Policy Directive (AFPD) 11-2, *Aircrew Operations*; AFPD 11-4, *Aviation Service*; and AFI 11-202 Volume 3, *General Flight Rules*. It establishes the minimum Air Force operations procedures for personnel performing duties in the F-15. It applies to all F-15 A-D units, including the Regular Air Force, Air Force Reserve and Air National Guard (ANG). This Manual requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by Department of Defense Directive (DoDD) 5400.11, *DoD Privacy Program*. The applicable SORN(s) F011 AF XO A is available at: **http://dpclo.defense.gov/Privacy/SORNs.aspx**. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Instruction (AFI) 33-322, *Records Management and Information Management Governance Program*, and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility (OPR) listed above for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the OPR listed above using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate chain of command. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force. This publication incorporates Air Combat Command (ACC), ANG, Pacific Air Forces (PACAF) and United States Air Forces in Europe/Air Forces Africa (USAFE/AFAFRICA) supplements using the paragraph supplementation method. See **paragraph 1.4** for guidance on submitting comments and suggesting improvements. The

authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Instruction (DAFI) 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, as further identified in **paragraph 1.3** for non-tiered compliance items. **NOTE:** This instruction contains references to field (subordinate level) publications and forms which, until converted to departmental level publications and forms, may be obtained from the respective Major Command (MAJCOM) publication distribution office. Compliance with the attachments of this publication is mandatory. **NOTE:** The requirements of this publication are for default mission planning purposes. Nothing in this manual should be construed as overriding a crew member's judgment due to unforeseen operational necessity or other emergency circumstances.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. Paragraphs have been reorganized to increase standardization with other AFMAN 11-2 mission design series (MDS) Vol 3s and improve logical flow. Aircrew are required to thoroughly review this publication to understand the implications of the reorganization.

Chapter 1—GENERAL GUIDANCE **7**

1.1. Roles and Responsibilities. 7

1.2. Waivers. 7

1.3. Deviations. 7

1.4. Processing Changes. 8

Chapter 2—MISSION PLANNING **9**

2.1. Responsibilities. 9

2.2. General Procedures. 9

2.3. Map/Chart Preparation. 9

2.4. Briefing/Debriefing. 11

2.5. Unit Developed Checklists and Local Pilot Aids 12

2.6. Stowing Equipment in Bay 5. 13

2.7. Stowing Equipment on the Canopy Deck (Turtle Deck). 13

2.8. G-suit Use. 14

2.9. Chemical, Biological, Radiological, Nuclear, and High Yield Explosive (CBRNE) Operations. 14

Chapter 3—NORMAL OPERATING PROCEDURES 16

3.1. Ground Communications. 16
3.2. Ground Visual Signals. 16
3.3. Preflight. 16
3.4. Ground Operations. 17
3.5. Flight Lineup. 17
3.6. Before Takeoff Checks. 17
3.7. Takeoff. 17
3.8. Join-up/Rejoin. 18
3.9. Formation, General. 18
3.10. Tactical Formation. 19
3.11. Chase Formation. 20
3.12. Show Formation. 21
3.13. Maneuvering Parameters. 21
3.14. Ops checks. 22
3.15. Radio Procedures. 23
3.16. Change of Aircraft Control. 24
3.17. General Low Altitude Procedures. 24
3.18. Air Refueling. 26
3.19. Night Procedures. 26
3.20. Night Vision Goggles (NVGs) Procedures. 27
3.21. Fuel Requirements. 28
3.22. Approaches and Landings. 29
3.23. Overhead Traffic Patterns. 29
3.24. Tactical Overhead Traffic Patterns. 29
3.25. Touch-and-Go Landings. 30
3.26. Low Approaches. 30
3.27. Closed Traffic Patterns. 30
3.28. Back Seat Approaches and Landings. 30

3.29. Refer to Table 3.1. 31
Table 3.1. Weather Minimum Summary (Feet/NM/KM). 31
3.30. Refer to Table 3.2. 31
Table 3.2. Minimum Altitude Summary (in Feet AGL). 31
Chapter 4—INSTRUMENT PROCEDURES **32**
4.1. Approach Category. 32
4.2. Takeoff and Join-up. 33
4.3. Trail Procedures. 33
4.4. Formation Break-up. 36
4.5. Formation Penetration. 36
4.6. Formation Approach. 36
4.7. Simulated Instrument Flight. 36
4.8. Use of the Heads Up Display (HUD). 36
4.9. Use of the Joint Helmet Mounted Cueing System (JHMCS). 36
Chapter 5—AIR-TO-AIR WEAPONS EMPLOYMENT **37**
5.1. References. 37
5.2. Simulated Gun Employment. 37
5.3. Maneuvering Limitations. 37
5.4. Aerial Gunnery Tow Procedures. 37
Chapter 6—AIR-TO-SURFACE WEAPONS EMPLOYMENT **40**
6.1. General. 40
6.2. Weapons Delivery Parameters. 40
6.3. Simulated Attacks against Range, Off-Range or Manned Targets: 40
Chapter 7—ABNORMAL OPERATING PROCEDURES **41**
7.1. General. 41
7.2. Ground Aborts. 41
7.3. Takeoff Aborts. 41
7.4. Air Aborts. 42
7.5. Engine Malfunctions. 43

7.6. Radio Failure. 43
7.7. Severe Weather Penetration. 44
7.8. Lost Wingman Procedures. 44
7.9. Spatial Disorientation (SD). 45
7.10. Armament System Malfunctions. 46
7.11. Post Arresting Gear Engagement Procedures. 47
7.12. In-flight Practice of Emergency Procedures. 47
7.13. Search and Rescue Combat Air Patrol (SARCAP) Procedures. 48
7.14. Lateral Asymmetry. 49
Table 7.1. Asymmetry ROTs. 49
Chapter 8—LOCAL OPERATING PROCEDURES 50
8.1. This chapter is reserved for unit local operating procedures. 50
8.2. Procedures. 51
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 53
Attachment 2—GENERAL BRIEFING GUIDE 60
Attachment 3—SPECIAL SUBJECT BRIEFING GUIDE 62
Attachment 4—ADVANCED HANDLING/INSTRUMENT BRIEFING GUIDE 63
Attachment 5—AIR REFUELING BRIEFING GUIDE 64
Attachment 6—AIR COMBAT TRAINING (ACBT) / INTERCEPT BRIEFING GUIDE 66
Attachment 7—ESCORT MISSION BRIEFING GUIDE 70
Attachment 8—AERIAL GUNNERY TOW COORDINATION BRIEFING GUIDE 72
Attachment 9—AERIAL GUNNERY BRIEFING GUIDE 74
Attachment 10—LOW-LEVEL NAVIGATION BRIEFING GUIDE 76
Attachment 11—NVGS BRIEFING GUIDE 78
Attachment 12—ALERT BRIEFING GUIDE 81
Attachment 13—HIGH ANGLE STRAFE (HAS) BRIEFING GUIDE 84
Attachment 14—MARITIME AIR SUPPORT (MAS) AND BASIC SURFACE ATTACK (BSA) BRIEFING GUIDE 87

Attachment 15—CREW COORDINATION/PASSENGER/GROUND CREW BRIEFING GUIDE **90**

Attachment 16—MISSION DEBRIEFING GUIDE **92**

Chapter 1

GENERAL GUIDANCE

1.1. Roles and Responsibilities.

1.1.1. This instruction, in conjunction with other governing directives, prescribes procedures for operating F-15 aircraft under most circumstances. It is not a substitute for sound judgment.

1.1.2. Commanders. Commanders at their respective Tier levels are responsible for complying with guidance in this manual. F-15E flying unit Wing Commanders, delegated no lower than the Operations Group (OG) Commander (or equivalent), are responsible for providing local operating guidance to supplement the requirements in **Chapter 8** of this manual. **(T-2)**.

1.1.3. Pilot in Command (PIC) Authority. The PIC, regardless or rank, is responsible for, and is the final authority for the operation of the F-15E aircraft. Pilots will use best judgement to safely conduct flying operations.

1.1.4. Supplements. Comply with applicable supplements to all guidance reference in this manual. Develop additional supplements in accordance with DAFI 33-360.

1.1.5. Procedures not specifically addressed may be accomplished if they enhance safe and effective mission accomplishment.

1.2. Waivers.

1.2.1. Forward waiver requests through appropriate channels to the applicable MAJCOM/A3 or equivalent, or Commander, Air Force Forces (COMAFFOR) for those aircrew and assets under the COMAFFOR's oversight, for approval. The COMAFFOR or MAJCOM/A3 (or equivalent) will notify ACC/A3 of waivers within 72 hours of issuance. In addition to the waiver processing requirements prescribed by DAFI 33-360, wing commanders will notify the publication OPR within 72 hours of waiver approval. **(T-1)**. In accordance with DAFI 33-360, a copy of the approved waiver must follow within 30 days of issuance. An email to the waived publication OPR that includes a completed AF Form 679, *Air Force Publication Compliance Item Waiver Request/Approval* or equivalent will suffice. Waiver authority for supplemental guidance will be as specified in the supplement and approved through higher level coordination authority.

1.2.2. Comply with applicable supplements to all guidance referenced in this manual. Develop additional supplements in accordance with DAFI 33-360.

1.2.3. COMAFFOR staff will notify HQ ACC/A3 and home station MAJCOM/A3 of waivers within 72 hours of approval. **(T-2)**.

1.3. Deviations.

1.3.1. In the case of an urgent requirement or aircraft emergency, the pilot in command will take appropriate action(s) to safely recover the aircraft. **(T-2)**.

1.3.2. If time permits, specific MAJCOM/A3 approval will be obtained for one-time deviations from these procedures. **(T-2)**.

1.3.3. COMAFFOR staff will notify HQ ACC/A3 and home station MAJCOM/A3 of waivers within 72 hours of approval. **(T-2)**.

1.4. Processing Changes.

1.4.1. Submit recommended changes and questions about this publication through MAJCOM channels to the OPR per AFI 11-215, *USAF Flight Manuals Program* using AF Form 847.

1.4.2. The submitting MAJCOM will forward information copies of AF Forms 847 to all other MAJCOMs that use this publication. **(T-1)**. Using MAJCOMs will forward comments on AF Forms 847 to the OPR. **(T-1)**.

1.4.3. OPR will:

1.4.3.1. Coordinate all changes to the basic volume with affected MAJCOM/A3s. **(T-1)**.

1.4.3.2. Forward change recommendations to HQ AFFSA for staffing and HQ USAF/A3T approval. **(T-1)**.

Chapter 2

MISSION PLANNING

2.1. Responsibilities. The responsibility for mission planning is shared jointly by all flight members and the operations and intelligence functions of fighter organizations.

2.2. General Procedures.

2.2.1. Accomplish sufficient flight planning to ensure safe mission accomplishment to include fuel requirements, map preparation, and takeoff/landing data.

2.2.1.1. **(PACAF)** Fly planned over-water flights outside the local training area (i.e., deployments, cross countries, Program Depot Maintenance (PDM) inputs) as a two-ship minimum. OG/CC may approve single-ship missions.

2.2.1.2. **(USAFE/AFAFRICA)** In addition, when applicable, pilots will consult the following for mission planning:

2.2.1.2.1. Department of Defense (DoD) Flight Information Publications (FLIP) Planning Europe, Africa, and Middle East. **(T-2)**.

2.2.1.2.2. United Kingdom (UK) Military Low Fly Handbook. **(T-2)**.

2.2.1.2.3. UK Low Fly Notices to Airmen (NOTAMS). **(T-2)**.

2.2.1.2.4. UK Royal Flights. **(T-2)**.

2.2.1.2.5. AFI 11-202 Volume 3 USAFE SUP I, **Attachment 6**, *Control of Fighter Aircraft for Off Station Sorties/Divert*. **(T-2)**.

2.2.1.2.6. ASRR, *Airfield Suitability Report/Summary Airfield Restrictions*. **(T-2)**.

2.2.1.2.7. AFMAN 13-212 Volume 1, *Range Planning and Operations.* **(T-2)**.

2.2.1.2.8. AFPAM 11-216, *Air Navigation.* **(T-2)**.

2.2.2. Fuel Conservation. Design procedures for optimal fuel use and efficiency throughout all phases of mission execution. Incorporate enroute tasks to maximize use of airborne training opportunities.

2.3. Map/Chart Preparation.

2.3.1. Local Area Maps. A local area map is not required if the unit pilot aid includes jettison areas, divert information, controlled bailout areas, and provides a local area map of sufficient detail to remain within assigned training areas.

2.3.1.1. **(USAFE/AFAFRICA)** On flights from a deployed location, each aircrew will have available a local map annotated with designated flying areas, emergency airfields, buffer zones, control zones, and restricted or danger areas if this information is not available in the aircrew aid. **(T-2)**.

2.3.2. Charts. FLIP enroute charts may be used instead of maps on navigational flights within areas that are adequately covered by these charts.

2.3.2.1. **(USAFE/AFAFRICA)** Pilots may use UK Enroute Low Altitude Charts to supplement United States (U.S). DoD Enroute Low charts as deemed necessary. Aircrew will reference UK Flight Information Handbook for index and chart definitions contained on UK Enroute Low chart. **(T-2)**.

2.3.3. Low Altitude Maps.

2.3.3.1. Reference AFMAN 11-202V3, *Flight Operations*, for low altitude map requirements and procedures.

2.3.3.2. On low altitude flights, each aircraft in the flight will contain a current map (updated using Chart Update Manual or electronic equivalent) of the route/operating area. **(T-1)**. Pilots will ensure it is of such scale and quality that the detail of terrain features, hazards, and chart annotations permits navigation and safe mission accomplishment. **(T-1)**. Pilots will circle/highlight manmade obstacles at above planned flight altitude within 5 nautical miles (NM) of the planned route. **(T-1)**. Pilots will annotate time or distance tick marks and headings. **(T-1)**.

2.3.3.3. Highlight all manmade obstacles at or above the planned flight altitude.

2.3.3.4. In accordance with AFMAN 11-217, *Flight Operations*, annotate all maps with both an emergency route abort altitude (ERAA) for the overall route/area and minimum safe altitudes (MSA) for each leg of the intended route of flight.

2.3.3.4.1. Compute the ERAA (a.k.a. Emergency Safe Altitude (ESA)) in accordance with AFMAN 11-202V3.

2.3.3.4.2. Compute the MSA at a minimum of 1000 feet above the highest obstacle/terrain (rounded up to the next 100 feet) within 5NM of the planned course to include the aircraft turn radius.

2.3.3.5. [Continental United States (CONUS) Only] Aircrew members flying under visual flight rules (VFR) or inside Military Training Routes (MTR)s will supplement existing mission planning materials (e.g., FLIP, Area Planning Military Training Route Charts (AP/1B)) in accordance with AFI 11-202 Volume 3, Air Combat Command Supplement (ACCSUP), Chapter 2, mission planning requirements. **(T-2)**.

2.3.3.6. Low level charts and route books used during flight will be annotated with location and dimensions of class B/C/D airspace, civil/military airfields and other potential high density traffic areas (e.g., parachute activity areas and ultralight/hang glider/glider sites) within 5 NM of any planned VFR route or MTR lateral boundary. **(T-2)**. Applicable airfield approach control frequencies in the vicinity of class B/C/D airspace will be annotated and briefed on all such flights. **(T-2)**. In addition, annotate and brief the intersection of other Visual Route/ Instrument Route (VR/IR) routes (if applicable) and any other possible areas of conflict. **(T-2)**.

2.3.3.7. Use of sectional aeronautical charts in flight is not required.

2.3.3.8. Aircrew members flying outside CONUS will follow gaining MAJCOM, theater or host nation guidance on mission planning. **(T-2)**. If no gaining MAJCOM, theater or host nation guidance exists, use the best charts or Falcon View overlay options available to accomplish the requirements of **paragraph 2.3.3.6**

2.4. Briefing/Debriefing.

2.4.1. Briefings. Flight leaders are responsible for presenting a logical briefing that will promote safe, effective mission accomplishment. **(T-2)**. In addition to items required by AFI 11-202 Volume 3, all briefings will include the following:

2.4.1.1. All pilots/crewmembers/passengers must attend the flight briefing unless previously coordinated with unit supervisors. **(T-2)**. Anyone not attending the flight brief must receive a briefing on mission events and emergency procedures. **(T-2)**.

2.4.1.2. Ensure brief start time provides adequate time to discuss required items and accounts for mission complexity. **(T-2)**. As a minimum, begin briefs at least 1.5 hours before scheduled takeoff. Alert briefs will start in sufficient time to be completed prior to aircrew changeover. **(T-2)**.

2.4.1.3. Structure flight briefing to accommodate the capabilities of each flight member. **(T-2)**.

2.4.1.4. Briefers will reference applicable portions of briefing guides. **(T-2)**. Items listed may be briefed in any sequence. Those items published in AFIs; Air Force Tactics, Techniques, and Procedures (AFTTP) manuals or unit standards and understood by all participants may be briefed as "standard." Specific items not pertinent to the mission do not need to be covered.

2.4.1.5. Takeoff and landing data (TOLD) will be annotated on mission data cards. **(T-2)**. The minimum TOLD required is maximum abort speed (dry/wet), takeoff distance and speed, single engine takeoff speed (SETOS), and normal/heavy weight landing distance (dry/wet).

2.4.1.6. Review takeoff data, and ensure every member of the flight understands it. Place particular emphasis on takeoff and abort factors during abnormal situations such as short/wet runway, heavy gross weights, non-standard cable configurations, and abort sequence in formation flights.

2.4.1.7. Ensure contracts, roles, and responsibilities of each flight member are established, briefed, and debriefed.

2.4.1.8. Include mission priorities, significant rules (e.g., Rules of Engagement (ROE), Special Instructions (SPINS), Training Rules (TRs)), task management, weather, NOTAMs, and Emergency Procedures (EPs).

2.4.1.9. Ensure a formation deconfliction, blind, and get well plan for every phase of flight is briefed and every flight member understands the plan (see **paragraph 3.10**). All flight members are responsible for executing this plan.

2.4.1.10. When dissimilar aircraft are flown in formation, brief flight responsibilities, proper formation position (to ensure adequate wingtip clearance), and aircraft-unique requirements for each phase of flight.

2.4.1.11. For missions using night vision goggles (NVGs), emphasize proper tuning, use, and limitations. See **paragraph 3.20** for additional procedures.

2.4.1.12. For all low-level mission briefings, place emphasis on obstacle/ground avoidance, altitude-warning features (low altitude warning tone) low altitude comfort level, and complacency avoidance.

2.4.1.13. Brief an appropriate alternate mission for each flight. The alternate mission must be less complex than the primary unless wings have developed and published standard alternate mission profiles. **(T-2)**. The one exception to this rule is that air combat maneuvers (ACMs) may be briefed as an alternate mission from a basic fighter maneuvers (BFMs) surge brief. Brief specific mission elements that differ from the primary mission or published profiles. Mission elements or events may be modified and briefed airborne as long as flight safety is not compromised. Flight leads will ensure all flight members acknowledge changes. **(T-2)**. Do not fly unbriefed missions or events.

2.4.1.14. All missions will be debriefed. **(T-2)**.

2.4.2. **Deployed Operations, Exercise, and Quick Turn Briefings.** If all flight members attend an initial or mass flight briefing, the flight lead on subsequent flights need brief only those items that have changed from the previous flight(s).

2.4.2.1. **(PACAF, USAFE/AFAFRICA)** On multiple-go days when aircraft turn times do not allow follow-on mission brief(s) and only the initial flight brief is accomplished for all goes, the following guidance applies:

2.4.2.1.1. **(PACAF, USAFE/AFAFRICA)** Upgrade missions will be planned for the first sortie flown. **(T-3)**. Subsequent missions will be of equal or less complexity with no upgrade training planned without OG/CC approval. **(T-2)**.

2.4.2.1.2. **(PACAF, USAFE/AFAFRICA)** Participants in continuation training missions may fly their primary or alternate missions in any sequence.

2.4.3. Briefing Guides. Mission briefing guides are contained in the attachments. Units may augment these guides as necessary. Pending development by a higher headquarters, units that fly missions not covered by this instruction or its supplements will develop briefing guides for those missions and submit them to MAJCOM/A3TO for review. **(T-2)**.

2.5. Unit Developed Checklists and Local Pilot Aids.

2.5.1. Unit developed checklists may be used in lieu of flight manual checklists provided they contain, as a minimum, all items (verbatim and in order) listed in the applicable checklist.

2.5.2. Units will produce a pilot aid that, as a minimum, includes:

2.5.2.1. Briefing guides. **(T-2)**.

2.5.2.2. Local radio channelization and airfield diagrams. **(T-2)**.

2.5.2.3. Emergency information (impoundment procedures, emergency action checklists, no radio (NORDO) procedures, divert information, search and rescue (SAR) procedures, and on-scene commander (OSC) checklist). **(T-2)**.

2.5.2.4. Arresting gear information for divert bases. **(T-2)**.

2.5.2.5. Bailout and jettison areas. **(T-2)**.

2.5.2.6. Cross-country procedures to include: command and control, engine documentation, (Joint Oil Analysis Program) JOAP samples, and aircraft servicing. **(T-2)**.

2.5.2.7. Other information as deemed necessary by the units. For example: stereo flight plans, turnaround procedures, local training areas, instrument preflight, and alert procedures. **(T-2)**.

2.6. Stowing Equipment in Bay 5.

2.6.1. Stow containers or baggage with hard sides inside Bay 5. **(T-2)**. Stow only locally manufactured fabric intake covers and soft-sided personal equipment baggage, such as hang-up or A-3 bags, in Bay 5. **(T-2)**. Items stowed in Bay 5 will be:

2.6.1.1. Positioned 1 foot aft of the top of the canopy thruster "catcher's mitt." **(T-2)**.

2.6.1.2. Forward of the trailing edge of the aft circuit breaker panel on the right wall of Bay 5 (do not interfere with internal countermeasures set cables). **(T-2)**.

2.6.1.3. Below the top of the circuit breaker panels on the right wall of Bay 5. **(T-2)**.

2.6.1.4. Secured with non-stretchable cord in such a manner to prevent movement in all three axes of motion. Place aircraft safety equipment and egress pins in locally manufactured fabric bags and stow in panels 154L, 35, 47L and R or in the cockpit map case. The pilot is responsible for ensuring items stowed in Bay 5 are loaded correctly and properly secured. This guidance does not preclude the pilot from taking any action necessary for safety of flight. Pilots will comply with the following restrictions: **(T-2)**.

2.6.1.4.1. Aircraft with items stowed in Bay 5 are restricted from performing aerobatics. **(T-2)**.

2.6.1.4.2. Training missions in aircraft with items stowed in Bay 5 are restricted to limited maneuvering training rules. **(T-2)**. At no time during the flight will the pilot execute zero or negative gravitational load factor (G) maneuvers unless safety of flight dictates. **(T-2)**.

2.7. Stowing Equipment on the Canopy Deck (Turtle Deck).

2.7.1. Stowing any items on the canopy deck during flight is prohibited. **(T-2)**. Failure to follow this procedure could cause complications during ejection to include:

2.7.1.1. Impeding ejection and/or canopy motion during jettison/ejection

2.7.1.2. Obstructing airflow to the seat-mounted pitot tubes

2.7.1.3. Obstructing parachute deployment from the seat

2.7.1.4. Affecting performance/trajectory/stability of the seat and/or jettisoned canopy

2.7.1.5. Entangling the seat, canopy, and/or aircrew to each other

2.7.1.6. Damaging the ejections seat and/or canopy.

2.8. G-suit Use. Pilots will wear anti-G garments on all flights regardless of anticipated G's. **(T-2)**.

2.9. Chemical, Biological, Radiological, Nuclear, and High Yield Explosive (CBRNE) Operations.

2.9.1. **General Information.** Potential adversary use of CBRNE weapons against a friendly airfield presents a serious threat to flying operations. Although the most effective way for aircrews to avoid this threat is to be airborne before these weapons are detonated or dispersed and then land at a field that has not been contaminated, all personnel must be prepared to operate from a field that has come under CBRNE attack. **(T-2)**.

2.9.2. **Mission Preparation.** Be aware of the status of the CBRNE environment at the planned launch and recovery airfields, potential divert bases, and throughout the area in which the sortie may fly. Know the current and forecast surface wind direction and on the Mission Oriented Protective Posture (MOPP) level in effect for relevant sectors of the airfield. Don appropriate aircrew chemical defense equipment (ACDE) or ground crew ensemble (GCE) to match the appropriate MOPP level (reference AFTTP 3-4, *Airman's Manual*) and carry individual protective equipment (IPE) as required.

2.9.3. **Stepping to Fly and Aircraft Preflight.** This may entail donning ACDE or transitioning from GCE to ACDE. Take precautions to protect aircrew from injury and or contamination while in transit from the squadron facility to the aircraft. If possible, transport aircrew in a vehicle that provides overhead cover (enclosed vehicle). If aircrew travel on foot is unavoidable, choose a route that takes maximum advantage of available overhead cover (sun shades, buildings, etc.) to avoid agents that may be settling from the air. If extra aircrew members are available for preflight duties, consider assigning them to do so wearing GCE. This will allow the aircrew actually flying to minimize exposure.

2.9.3.1. **Alarm Red (or theater equivalent) Prior to Engine Start.** If Alarm Red occurs during the step or preflight process, take cover and don appropriate MOPP. This may require use of the ground crew mask. A hardened aircraft shelter (HAS) provides optimum protection, if available. Use caution if entering a HAS that contains aircraft or equipment. Close doors after entry. If a HAS or other overhead cover is not immediately available, accept the best rapidly reachable cover.

2.9.4. **Engine Start to Takeoff.** If a HAS is available, use it to minimize exposure time by accomplishing aircraft arming and end of runway (EOR) procedures inside the HAS (if local procedures permit) and by delaying taxi time as long as possible prior to takeoff.

2.9.4.1. **Aircraft Launch to Survive.** Units will develop local procedures to provide this option to the commander. **(T-2)**. In general, aircraft may launch to survive any time after engine start if they have sufficient fuel and safe, expeditious access to a runway. This option may only be practical for aircraft that are near the EOR prior to takeoff or that have just landed.

2.9.4.2. **Alarm Red (or theater equivalent) Prior to Taxi.** If in a HAS, the normal procedure is to shut down. Ensure ground personnel are aware of the alarm warning, as engine noise may preclude effectiveness of normal alert notification procedures. Use hand signals if necessary to be sure that ground personnel assume proper MOPP and close HAS doors. If not in a HAS, procedures may include launch to survive.

2.9.4.3. **Alarm Red (or theater equivalent) After Taxi.** Units typically establish procedures for this contingency depending on whether additional protection is available along the taxi route. For instance, if empty HAS is available, taxiing aircraft may be directed to shelter there. Ideally, ground crew sheltering in such a HAS would be available to assist in normal engine shutdown procedures and to close HAS doors. If protection is not available, the best option may be launch to survive. Maintain contact with command and control (C2) entities (wing operations center, maintenance operations center, supervisor of flying, etc.) to ensure unity of effort in the overall plan.

2.9.5. **Takeoff to Landing.**

2.9.5.1. **Contamination.** If chemical warfare (CW) agent contamination occurs prior to takeoff, flying the aircraft will dissipate the agent to some degree. The greatest dissipation will occur during flights at lower altitudes and longer airborne times. Because the agent may have entered wheel wells, flaps, etc., consider flying in landing configuration to increase airflow to these areas. However, merely flying the aircraft is unlikely to achieve complete decontamination.

2.9.5.2. **Preparing to Land.** Aircrew should remain aware of the status of primary and alternate landing locations. Do not attempt to land during Alarm Red situations unless there is no other option. Follow C2 directions and either hold or divert. If mission needs prevent divert, hold until the Alarm Red (or theater equivalent) has cleared or become an Alarm Black. Prior to landing, gain awareness of contaminated sectors of the airfield and of current/forecast surface winds. Use this information in conjunction with C2 direction to plan a route from landing to engine shutdown. The liquid deposition phase following a CW airburst attack can extend up to 1 hour. If landing during Alarm Black, expect a contaminated environment and MOPP 4.

2.9.6. **Landing to Engine Shutdown.** Take advantage of any protection available, minimizing taxi time and distance. Maintain contact with C2 in order to remain aware of unexploded ordnance and/or damage to airfield movement surfaces. If a HAS is available and local procedures permit, accomplish aircraft de-arm and EOR procedures there. If Alarm Red (or theater equivalent) occurs between landing and engine shutdown, considerations are similar to those in **paragraph A2.4**

2.9.7. **After Engine Shutdown.** Don appropriate MOPP. **(T-2)**. If circumstances permit, accomplish normal post-flight inspection procedures. **(T-2)**. If the aircraft is not contaminated, close the canopy. If there is any suspicion of personnel contamination, aircrew will process through an aircrew contamination control area. **(T-2)**. Accomplish maintenance debriefings under cover to the maximum extent possible.

Chapter 3

NORMAL OPERATING PROCEDURES

3.1. Ground Communications.

3.1.1. The pilot will accomplish the ground crew briefing (when required) in accordance with the briefing guide contained in this instruction. **(T-2)**.

3.1.2. Prior to starting the jet fuel starter (JFS) in a B/D model with the rear cockpit (RCP) occupied, the pilot will get an "OK" signal from the rear cockpit occupant. **(T-2)**.

3.1.3. Normally, the pilot and ground crew will communicate using the intercom during all engine start and pre-taxi checks. **(T-2)**. Use the intercom system, to the maximum extent possible, anytime aircraft engines are operating and maintenance technicians are performing tasks on the aircraft. Units with active air defense commitments may waive the use of intercom during alert scrambles. During the after engine start flight control checks, confirm the proper movement and position of the flight control surfaces with the crew chief.

3.2. Ground Visual Signals.

3.2.1. When ground intercom is not used, use visual signals in accordance with AFMAN 11-218, *Aircraft Operations and Movement on the Ground*, and this instruction. All signals pertaining to operation of aircraft systems will originate with the pilot. **(T-2)**. The crew chief will repeat the given signals when it is safe to operate the system. **(T-2)**. The pilot should not activate any system that could pose danger to the ground crew prior to receiving proper acknowledgment from ground personnel. The following signals augment AFMAN 11-218.

3.2.1.1. **JFS Start.** With clenched fist, pilot makes a pulling motion.

3.2.1.2. **Flight Controls Check.** Raise arm, clench fist, and make a stirring motion.

3.2.1.3. **Brake Check.** Hold left or right arm horizontal, open hand and push forward, breaking at the wrist (as in applying rudder pedal pressure with feet).

3.2.1.4. **Digital Electronic Engine Control (DEEC) Check.** With the fingers and thumb of each hand extended and joined at the tips, open and close the fingers and thumbs of both hands simultaneously, simulating nozzle opening and closing.

3.2.1.5. **Loss Of Brakes While Taxiing.** Lower tailhook.

3.2.1.6. **Gun Armament Check.** Point index finger forward with thumb upward simulating a pistol and shake head (yes or no).

3.3. Preflight.

3.3.1. Do not carry baggage/equipment in an unoccupied F-15B/D rear cockpit (**Exception**: forms/maps may be stowed in the map case).

3.3.2. In lieu of Higher Headquarters (HHQ) guidance (e.g., theater special instructions), set the emergency locator beacon to AUTO while on local training missions and ferry flights.

3.4. Ground Operations.

3.4.1. In accordance with AFTTP 3-3.F-15, the minimum taxi interval is 150 feet staggered or 300 feet in trail. Spacing may be reduced when holding short of or entering the runway.

3.4.2. Do not taxi during snow and/or icy conditions until the taxi route and runway have been checked for safe conditions. In this case, taxi on the centerline with a minimum of 300 feet spacing.

3.4.2.1. **(PACAF, USAFE/AFAFRICA)** Minimum runway condition reading/report (RCR) on taxiways for taxi operations is RCR 10.

3.4.3. Maximum taxi speed during sharp turns is 10 knots. Above 10 knots the aircraft may skid and/or depart the three-point attitude.

3.4.4. Quick Check and Arming. Keep hands in view of ground personnel during quick check, arming or de-arming operations. If the intercom system is not used during EOR checks, the pilot will establish and maintain visual contact with the ground personnel to allow the use of visual signals. **(T-2)**.

3.4.5. Do not taxi in front of aircraft being armed/de-armed with forward firing ordnance.

3.5. Flight Lineup. Flights will line up as appropriate based on weather conditions, runway conditions, and runway width. **(T-2)**. Use a minimum of 500 feet spacing between separated elements/flights. If runway width precludes line-up with wingtip clearance between all aircraft in the flight, use 500 feet spacing between elements or delay run-up until the preceding aircraft/element releases brakes.

3.6. Before Takeoff Checks. Just prior to takeoff, all flight members will inspect each other for proper configuration and any abnormalities. **(T-2)**. Wingmen will indicate they are ready for takeoff by a head nod, radio call, or landing/taxi light signal. **(T-2)**.

3.7. Takeoff. Per MAJCOM guidance, OG/CC may waive RCR minimum for specified units operating in cold weather locations, but in no case will takeoffs be conducted with an RCR of less than 8. **(T-2)**.

3.7.1. **(ACC/ANG)** ***Do not takeoff if the RCR is less than 12.*** **(T-2)** .

3.7.2. On training missions, do not takeoff if the computed takeoff roll exceeds 80 percent of the available runway. For single ship takeoffs, if the single ship computed military power takeoff distance exceeds one-half of the available runway, takeoff using afterburner.

3.7.3. When operating from airfields equipped with a compatible, remotely operated cable, ensure the departure end cable is raised for all takeoffs and landings, unless another departure end cable is in place.

3.7.4. Use a minimum of 10 seconds (15 seconds when using afterburners) takeoff interval between aircraft/elements. When joining “on top” use a minimum of 20 seconds takeoff interval.

3.7.5. Wing/group commander or operations group commander (SOF for ANG) may approve intersection takeoffs if operational requirements dictate.

3.7.6. Only perform formation takeoffs when required for safe launch of aircraft.

3.8. Join-up/Rejoin.

3.8.1. Day weather criteria for a VFR join-up underneath: ceiling 1500 feet and visibility 3 statute miles.

3.8.2. Reference AFTTP 3-3.F-15 for formation rejoin standard contracts (airspeeds, bank angle, and formation position).

3.8.3. **Battle Damage Checks.** When circumstances permit, flight leads will direct a battle damage check after each mission prior to or during return to base (RTB). **(T-2)**. Except at night/instrument meteorological conditions (IMC), this check is mandatory following the expenditure of any ordnance (including all types of 20mm ammunition). Fly no closer than fingertip formation spacing. Ensure deconfliction in accordance with AFTTP 3-3.F-15.

3.8.4. For further join-up procedures, reference AFTTP 3-3.F-15 and see Night Join-up **paragraph 3.19.3** and **Chapter 4.**

3.8.5. Air-to-Air Interrogator (AAI), Identification Friend or Foe/Selective Identification Feature (IFF/SIF) Operation.

3.8.5.1. Military use of AAI Mode 4 interferes with air traffic control (ATC) and civil Mode 3/C codes. F-15 AAI use in the National Airspace System (NAS) will comply with the following:

3.8.5.1.1. To reduce the potential for adverse effects on combat identification, ATC, and traffic collision avoidance systems, aircrew will limit interrogations to the minimum required for the mission. **(T-2)**.

3.8.5.1.2. For AAI operation within the United States and its possessions (US&P), aircrew will comply with the equipment and operational restrictions as specified in the applicable radio frequency authorization(s) which can be obtained from the unit spectrum management office. **(T-2)**.

3.8.5.1.3. Outside the Continental U.S. (OCONUS), follow gaining MAJCOM, theater, or host nation guidance for AAI operation. **(T-2)**.

3.9. Formation, General.

3.9.1. In IMC, the maximum flight size in close/route formation is four aircraft except when flying in close formation with a tanker.

3.9.2. Do not use rolling maneuvers to maintain or regain formation position below 5000 feet above ground level (AGL) or in airspace where aerobatics are prohibited. **(T-2)**.

3.9.3. Airborne visual signals will be in accordance with AFPAM 11-205, *Aircrew Quick Reference to Aircraft Cockpit and Formation Flight Signals*. **(T-2)**. For four-ship flights, formation changes will be initiated by radio call, when practicable. **(T-2)**. A radio call is mandatory when directing position changes at night or under instrument conditions, and all wingmen will acknowledge prior to initiating the change. **(T-2)**.

3.9.4. Flight leaders will not break up formations until each wingman has a positive fix from which to navigate (visual, radar, embedded global positioning system/inertial navigation system (EGI), or tactical air navigation (TACAN)). **(T-2)**.

3.9.5. Changing Leads.

3.9.5.1. When flying in limited visibility conditions, initiate lead changes from a stabilized, wings level attitude.

3.9.5.2. The minimum altitude for a lead change is 500 feet AGL over land or 1000 feet AGL over water. For night and IMC see **paragraph 3.17.8.2**

3.9.5.3. Do not initiate lead changes with the wingman further aft than 30 degrees from line abreast. **(T-2)**.

3.9.5.4. Flight/element leads will not initiate a lead change unless the aircraft assuming the lead is in visual contact and in a safe position to do so. **(T-2)**. When flying an instrument approach in IMC, do not initiate lead changes inside the final approach fix (FAF). **(T-2)**.

3.9.5.5. Initiate a lead change by visual signal or radio call.

3.9.5.6. Acknowledge receipt of the lead by head nod or radio call, as appropriate.

3.9.5.7. A lead change is effective upon acknowledgment.

3.9.5.8. The former leader then moves to the appropriate wing position.

3.9.6. The flight lead is always responsible for flight actions, regardless of the physical position in which he flies. Wingmen should always be prepared to fly the number one position if, in the judgment of the flight lead, such action is warranted.

3.9.7. Pilots who are not flight leads may lead limited portions of missions provided an instructor pilot (IP) or squadron supervisor is in the same element.

3.9.8. Dissimilar Formations. Dissimilar aircraft may be flown in the same formation if mission requirements dictate or to expedite traffic flow during departures and recoveries. Specific procedures will be thoroughly pre-briefed. **(T-2)**.

3.10. Tactical Formation.

3.10.1. General. In addition to guidance in AFTTP 3-3.F-15, apply the following rules for flight path deconfliction during tactical maneuvering:

3.10.1.1. Flight/element leads will consider wingman/element position and ability to safely perform a maneuver before directing it. **(T-2)**.

3.10.1.2. Wingmen/elements must maneuver relative to the flight lead/lead element and maintain sight. **(T-2)**. Trailing aircraft/elements are responsible for deconflicting with lead aircraft/elements.

3.10.1.3. Wingmen/elements will normally cross above the lead/lead element when deconfliction is required. **(T-2)**.

3.10.2. Loss of Visual. Use the following procedures when one or more flight members/elements lose visual contact within the formation:

3.10.2.1. When any flight member/element calls “Blind,” then the appropriate flight member/element will immediately respond with “Visual” and a position report or “Blind.” **(T-2)**.

3.10.2.2. When the other flight member/element is also "Blind," that pilot will transmit her or his altitude as an immediate action to deconflict flight paths. **(T-2)**. The flight lead will ensure a minimum of 500 feet altitude separation is maintained. **(T-2)**. Avoid climbs/descents through the deconfliction altitude when possible.

3.10.2.3. When there is not a timely acknowledgment of the original "Blind" call, then the flight member/element initiating the call will maneuver away from the last known position of the other flight member/element and alter altitude. **(T-2)**.

3.10.2.4. If visual contact is still not regained, the flight leader will take additional positive action to ensure flight path deconfliction within the flight to include a terminate/knock-it-off call if necessary. **(T-2)**. Consider scenario restrictions such as sanctuary altitudes and/or adversary blocks.

3.10.2.5. Aircraft will maintain altitude separation until a visual is regained **(T-2)**. If necessary, aircraft will navigate with altitude separation until mutual support is regained. **(T-2)**.

3.10.2.6. During certain tactical formations (tactical, fighter data link (FDL) tactical, night tactical, weather intercepts, wedge, etc.) visual may be lost due to environmental conditions, terrain, or desired tactics. Always ensure two sources of deconfliction are maintained until visual is re-established (e.g., Air-to-Air (A/A) TACAN, FDL, sanctuary altitudes, geographic deconfliction, radar situational awareness (SA), AAI).

3.10.3. Two-Ship. Deconfliction during tactical maneuvering of two-ship formations is in accordance with AFTTP 3-3.F-15.

3.10.4. Three/Four-Ship (or greater). When flights of more than two aircraft are in tactical formation, low altitude maneuvering trailing aircraft/element(s) will maintain sufficient spacing so that primary emphasis during formation maneuvering/turns is on low altitude awareness and deconfliction within elements, not on deconfliction between elements. **(T-2)**.

3.11. Chase Formation.

3.11.1. Restrictions. Any pilot may fly safety chase for aircraft under emergency or impending emergency conditions. In addition:

3.11.1.1. Qualified pilots (including initial qualification training (IQT)/mission qualification training (MQT) pilots who have successfully completed an Instrument/Qualification evaluation) may chase as safety observer for aircraft performing simulated instrument flight or hung ordnance patterns.

3.11.1.2. Specialized missions (e.g., operational test and evaluation (OT&E), weapon system evaluation program (WSEP), live weapons delivery) and training conducted in accordance with AFMAN 11-2F-15 Volume 1, *F-15 Aircrew Training*, may be chased by combat mission ready (CMR)/basic mission capable (BMC) pilots designated by group/squadron commanders.

3.11.1.3. All other chase events may only be flown by IP/ stan/eval flight examiners (SEFEs) or upgrading IPs under the supervision of an IP.

3.11.2. Procedures:

3.11.2.1. A safety observer in a chase aircraft, except IP/SEFE/specialized mission chase, will maneuver in a 30-60 degree cone with nose/tail clearance to 1000 feet, to effectively clear and/or provide assistance. **(T-2)**.

3.11.2.2. IP/SEFE/specialized mission aircraft will maneuver as necessary, but must maintain nose/tail separation until required to transition to close formation when deemed necessary by the IP/SEFE. **(T-2)**.

3.11.2.3. No chase aircraft will stack lower than the lead aircraft when below 1000 feet AGL **(T-2)**.

3.12. Show Formation. Brief and fly show formations as approved. **(T-2)**. Refer to AFI 11-209, *Participation in Aerial Events*, and applicable MAJCOM directives for specific rules and appropriate approval levels to participate in static displays and aerial events. **(T-2)**.

3.13. Maneuvering Parameters.

3.13.1. Minimum Altitudes. For aerobatics, remain above 5000 feet AGL. During nose high/low speed and air handling characteristics (AHC) vertical maneuvering ensure maneuvers are terminated to allow recovery above 5000 feet AGL.

3.13.2. Avoid flight through wingtip vortices and jetwash. If it is unavoidable, immediately unload the aircraft to approximately 1 G.

3.13.3. Do not extend flaps to improve aircraft maneuvering performance during air combat maneuvering (ACBT).

3.13.4. Long Term Structural Health Mitigation Measures.

3.13.4.1. In order to alleviate unnecessary structural fatigue on the F-15 fleet while making pilots aware of existing service life issues, units will comply with the following restrictions:

3.13.4.1.1. Limit maneuvering during administrative portions of the flight (to/from airspace, patterns, etc.) to less than 5Gs to the maximum extent possible. **(T-2)**.

3.13.4.1.2. Minimize high G, heavy weight maneuvering if not necessary to accomplish the mission. **(T-2)**.

3.13.4.1.3. Minimize unnecessary operation in flight regions most damaging to the F-15 structure to include:

3.13.4.1.3.1. Repetitive occurrences in double rate/solid tone overload warning system (OWS) beeper. **(T-2)**.

3.13.4.1.3.2. Consistent flight transients within the 6-7.5 G region. **(T-2)**.

3.13.4.2. Proper tactics and essential proficiency training are paramount to combat effectiveness and should not be sacrificed in an attempt to conserve structural health. **(T-2)**.

3.14. Ops checks.

3.14.1. Accomplish sufficient operations checks to ensure safe mission accomplishment. Additionally, each pilot should monitor the fuel system carefully throughout the flight to identify low fuel, trapped fuel, or out of balance situations as soon as possible. Ops checks are required:

3.14.1.1. During climb and at level off after takeoff.

3.14.1.2. Prior to each (D)ACBT engagement or intercept. In addition, a check for proper operation of all transfer tanks (wing tanks balanced and tank 1 feeding) will be performed prior to and between engagements or planned maneuvering above 30 units angle of attack (AOA). **(T-2)**.

3.14.1.3. Following Air Refueling.

3.14.2. Minimum items to check are engine instruments, total and internal fuel quantities/balance, G-suit connection, oxygen system (PRICE check), cabin altitude, pilot physiology, and G meter/OWS.

3.14.3. For formation flights, the flight leader may initiate ops checks by radio call or visual signal and wingmen will respond appropriately. **(T-2)**.

3.14.3.1. The query and response for ops checks will include pointer over counter readings (Example: "13.5 over 15.0"). **(T-2)**. **Exception**: Total fuel only may periodically be used during high demand phases of flight. Following external fuel consumption, ensure tank- one is feeding correctly and include "tank-one feeding" in the ops check. Add a "balanced" call to the normal Ops Check reply when wing fuel balance checks are required and the difference is no greater than 200 lbs. (Example: "8.5 squared and balanced"). If wingmen are within 500 lbs. of the flight lead, a "same" call may be used at the discretion of the OG/CC.

3.14.3.2. When more than one external tank is carried, add a "tanks feeding" call to the normal ops check reply. When flying the two-wing tank configuration, make a "tanks dry" call once the external tanks are confirmed dry and prior to transitioning to unlimited training rules. Once the "tanks dry" call has been made, no further reference to tanks need be made on subsequent Ops Checks.

3.14.3.3. When cabin pressure is on the correct schedule with regards to aircraft altitude, add a "normal" call to the normal ops check reply.

3.14.4. **G-awareness Exercise.**

3.14.4.1. Reference AFI 11-214, *Air Operations Rules and Procedures*, and AFTTP 3-3.F-15, Chapter 9. A G-awareness exercise will be briefed and accomplished on all tactical missions, day or night, with or without NVGs, anytime the plan includes or the likelihood to maneuver above 5 G's exists. **(T-2)**. Pilots must have sufficient visual cues to perform this maneuver. **(T-2)**. Flight members will maintain a minimum of 6000 feet separation between aircraft during the execution of all G-awareness exercises. **(T-2)**. Visual lookout and formation contracts will be the primary means for deconfliction. **(T-2)**. All available onboard systems will be used to aid in maintaining separation during execution. **(T-2)**. If poor weather or night illumination conditions preclude safe accomplishment of the G-awareness exercise, flight leads will modify the profile and

limit maneuvering accordingly. **(T-2)**. Briefings for night G-awareness maneuvers will emphasize wingman deconfliction procedures and maintaining spatial/situational awareness throughout the maneuver. **(T-2)**. In accordance with AFTTP 3-3.F-15C, both turns will be tactical 180's. **(T-2)**.

3.14.4.2. G-awareness maneuvers and all tactical portions of all missions will be taped hot mic for review of G straining techniques during the debrief. **(T-2)**.

3.14.4.3. **(PACAF, USAFE/AFAFRICA)** Do not use G-awareness turns for systems checks or other items that detract from the intended purpose.

3.14.4.4. If an operational requirement exists (as determined by the OG/CC) for an indication of individual aircraft stability, after the second G-awareness turn the pilot should accomplish a stability exercise once external wing tanks are empty and prior to unlimited maneuvering. This exercise will begin upon the flight lead's call at 350 KIAS and above 10,000 feet AGL (idle power for all configurations). **(T-2)**. Pilots will execute the stability exercise in accordance with AFTTP 3-3.F-15C, paragraph 9.3.1.3. **(T-2)**.

3.15. Radio Procedures.

3.15.1. Preface all communications with the complete flight call sign unless the tactical situation doesn't allow. Transmit only that information essential for mission accomplishment or safety of flight. Use visual signals when practicable.

3.15.2. Make a "knock-it-off" (KIO)" or "terminate" radio call to cease tactical maneuvering for any reason, particularly when a dangerous situation is developing. Any flight member may make this call. A KIO applies to any phase of flight and any type of mission. All participants will acknowledge a KIO by repeating the call. **(T-2)**.

3.15.3. The flight/mission leader will initiate all radio checks and channel changes. **(T-2)**.

3.15.4. Acknowledge radio checks that do not require the transmission of specific data by individual flight members in turn (example: "2, 3, 4"). Acknowledgment indicates the appropriate action is complete, in the process of being completed, or the flight member understands.

3.15.5. In addition to the radio procedures outlined in AFI 11-202 Volume 3, specific mission guides, and FLIP publications, the following radio transmissions are required:

3.15.5.1. All flight members will acknowledge understanding the initial ATC clearance. **(T-2)**. Acknowledge subsequent ATC instructions when directed by the flight lead, or anytime during trail departures/recoveries as detailed in **paragraph 4.3**

3.15.5.2. Gear Checks. Report gear down in accordance with AFI 11-202 Volume 3 after extending the landing gear and prior to the landing threshold. A wingman or chase need not make this call during a formation or chased approach.

3.15.6. Use brevity code and other terminology in accordance with AFI 11-214, AFTTP 3-1.1, *General Planning and Employment Considerations* and local standards.

3.16. Change of Aircraft Control.

3.16.1. Both pilots of an F-15B/D must know at all times who has control of the aircraft. Use the statement "You have the aircraft" to transfer aircraft control. **(T-2)**. The pilot receiving control of the aircraft will acknowledge "I have the aircraft." **(T-2)**. Once assuming control of the aircraft, maintain control until relinquishing it as stated above. **(T-2)**.

3.16.2. **Exception:** If the intercom fails, the pilot in the front cockpit (if not in control of the aircraft) will shake the stick and assume control of the aircraft, radios, and navigational equipment unless pre-briefed otherwise. **(T-2)**.

3.17. General Low Altitude Procedures.

3.17.1. Fly low level formation positions/tactics using AFTTP 3-1.F-15 and AFTTP 3-3.F-15, as guides.

3.17.2. During briefings, emphasize low altitude flight maneuvering and observation of terrain features/obstacles along the route of flight. For Low Altitude Training (LOWAT) over water/featureless terrain, include specific considerations with emphasis on minimum altitudes and spatial disorientation.

3.17.3. All obstacle avoidance planning will be based on MSA and ERAA as defined in **paragraph 2.3.3.3** **(T-2)**.

3.17.4. If unable to visually acquire or ensure lateral separation from known vertical obstructions which are a factor to the flight, flight leads will direct a climb no later than (NLT) 3 NM prior to the obstacle to ensure vertical separation. **(T-2)**. The climb must be to an altitude at or above briefed ERAA/MSA. **(T-2)**. Do not descend below this altitude until positional/situational awareness dictates it is safe to do so.

3.17.5. At altitudes below 1000 feet AGL, wingmen will not fly at a lower AGL altitude than lead. **(T-2)**.

3.17.6. When crossing high or hilly terrain, maintain positive G and do not exceed approximately 120 degrees of bank. Maneuvering at less than 1G is limited to upright bunting maneuvers.

3.17.7. The minimum airspeed for low level navigation is 300 KIAS unless operating in accordance with **paragraph 5.3.4**, low altitude restricted maneuvering target.

3.17.8. Minimum Altitudes. The unit commander will determine and certify pilots' minimum altitude in accordance with **AFMAN 11-2F-15 Volume 1,** as supplemented. **(T-2)**. Flight members participating in approved step-down training programs will comply with the requirements and restrictions of that program. **(T-2)**. The following minimum altitudes apply to low level training unless national rules, route restrictions, or a training syllabus specifies higher altitudes:

3.17.8.1. 500 feet AGL for: LOWAT Category I qualified pilots.

3.17.8.2. For night or IMC operation: The minimum altitude is 1000 feet above the highest obstacle within 5 NM of course unless operating under the conditions of **paragraph 3.21.3**

3.17.8.3. **(ACC/ANG)** 53 Wing, 57 Wing and AATC will fly low level as required for test sorties or in accordance with weapons school syllabi. **(T-3)**.

3.17.8.4. **(ANG/USAFE/AFAFRICA)** units participating in 53 WG and AATC tests will fly low level as required to meet test objectives. **(T-3)**.

3.17.8.5. For over water operation, the minimum altitude is 1000 feet above the surface unless in sight of land. If in sight of land, the minimum altitude may be lowered to 500 feet above the surface.

3.17.9. During all low altitude operations, the immediate reaction to task saturation, diverted attention, knock-it-off, or emergencies is to climb to a pre-briefed safe altitude (minimum 1000 feet AGL).

3.17.10. Weather minimums for visual low level training are 1500 feet ceiling and 3 miles visibility for any route or area, or as specified in FLIP for military training routes, unit regulations, or national rules, whichever is higher.

3.17.10.1. **(USAFE/AFAFRICA)** Weather minimums in countries where minimum low level altitude is 1000 feet AGL are: ceiling 2000 feet or 500 feet above planned flight altitude, whichever is higher, and visibility 8 Kilometers (KM). Minimums for low altitude intercept training in these countries are 2,500 feet AGL and 8KM visibility.

3.17.11. Low Level Route/Area Abort Procedures:

3.17.11.1. Compute and brief a low-level ERAA for all low level operations in accordance with **paragraph 2.3.3.4**

3.17.11.2. Visual Meteorological Conditions (VMC) route/area abort procedures:

3.17.11.2.1. Maintain safe separation from the terrain and other aircraft.

3.17.11.2.2. Comply with VFR altitude restrictions and squawk applicable IFF/SIF modes and codes.

3.17.11.2.3. Maintain VMC at all times. If unable, follow IMC procedures outlined below.

3.17.11.2.4. Attempt contact with controlling agency, if required.

3.17.11.3. IMC route/area abort procedures:

3.17.11.3.1. Immediately climb to, or above, the briefed ERAA.

3.17.11.3.2. Maintain preplanned ground track. Execute appropriate lost wingman procedures if necessary.

3.17.11.3.3. If deviations from normal route/area procedures are required, or if the ERAA/MSA is higher than the vertical limits of the route/area, squawk (IFF/SIF) emergency.

3.17.11.3.3.1. **(USAFE/AFAFRICA)** Squawk emergency or according to national rules for all IMC route aborts.

3.17.11.3.4. Attempt contact with the appropriate ATC agency for an instrument flight rules (IFR) clearance. If required to fly in IMC without an IFR clearance, cruise at appropriate VFR altitudes until IFR clearance is received.

3.17.11.3.4.1. **(USAFE/AFAFRICA)** Request IFR traffic separation if VMC cannot be maintained.

3.18. Air Refueling.

3.18.1. Pilots undergoing initial or recurrency training in air refueling will not refuel with a student boom operator (does not apply to KC-10). **(T-2)**.

3.18.2. Pilots will inform boom operator when refueling from particular tanker type (e.g., KC-10, KC-135, KC-46A) for the first time. **(T-2)**.

3.19. Night Procedures.

3.19.1. Night Ground Operations.

3.19.1.1. When ground personnel are working under the aircraft, the anti-collision lights should be OFF, and the position lights STEADY.

3.19.1.2. Taxi on the taxiway centerline with a minimum of 300 feet spacing.

3.19.1.3. Use the taxi light while taxiing unless it might interfere with an aircraft landing or taking off. The taxiing aircraft will come to a stop if the area cannot be visually cleared without the taxi light. **(T-2)**.

3.19.1.4. Both wingtip position lights, both wing root anti-collision lights, and 2 out of 3 formation lights per side, must be operational for flight (substituting a formation light in lieu of a wingtip position light is not permitted). **(T-2)**.

3.19.2. Night Takeoff. Following takeoff, each aircraft/element will climb on runway heading to 1000 feet AGL before initiating turns, except where departure instructions specifically preclude compliance. **(T-2)**.

3.19.3. Night Join-up. Weather criteria for night join-up underneath is a ceiling of 3000 feet and 5 miles visibility. After join-up, turn all the anti-collision lights OFF and position lights to STEADY except for the last aircraft in formation, which will keep the anti-collision light ON unless otherwise directed by the flight lead. **(T-2)**.

3.19.4. Night Formation Procedures.

3.19.4.1. When in positions other than fingertip, route, or tactical, maintain aircraft spacing primarily by instruments, FDL, Radar/AAI, and/or timing. "Use visual references as a secondary means of deconfliction. If aircraft spacing cannot be ensured, then establish altitude separation (1000 feet minimum). Crosscheck instruments at all times to ensure ground clearance.

3.19.4.2. Do not change lead or wing formation positions below 1500 feet AGL unless on radar downwind. Direct lead and position changes using the radio and from a stabilized, wings-level attitude.

3.19.4.3. Night Fingertip/Route Position. Night references for fingertip and route formation positions are specified in AFTTP 3-3.F-15.

3.19.5. Night Break-up. Prior to a formation break-up at night, the flight leader will transmit attitude, altitude, airspeed, and altimeter setting, which will be acknowledged by wingmen. **(T-2)**. Wingmen will confirm good navigational aids and return aircraft lighting, if reduced, to normal. **(T-2)**.

3.19.6. Night Landing. Normally land from an instrument straight-in approach. Refer to AFI 11-202 Volume 3, as supplemented, for specific procedures.

3.20. Night Vision Goggles (NVGs) Procedures.

3.20.1. USAF/MAJCOM guidance (including AFI 11-202 Volume 3 and AFI 11-214) outline NVGs procedures. AFTTP 3-1.F-15 and AFTTP 3-3.F-15 incorporate expanded tactical guidance. Additionally:

3.20.1.1. NVGs will only be worn by qualified flight members or when upgrading with NVGs with a qualified NVGs instructor in the flight. **(T-2)**.

3.20.1.2. Fly with NVGs only in MAJCOM approved NVGs-compatible lighted cockpits. **(T-2)**. Permanently modified NVGs-compatible cockpits that have a degraded light source may be used for NVGs missions at the discretion of the unit commander. All control and performance instruments must be illuminated by an NVGs-compatible light source to provide immediate reference. **(T-2)**.

3.20.1.3. Flight leads will brief the appropriate time to don/doff goggles for the sortie to be flown. **(T-2)**. Pilots will ensure deconfliction while donning/doffing goggles. **(T-2)**.

3.20.1.4. Qualified pilots may wear NVGs during ground operations if required to taxi in low illumination environments with Squadron/CC approval. Normal taxi clearances from obstacles still apply. In no case will NVGs be worn for takeoff or landing. **(T-3)**.

3.20.2. NVGs must be preflight tested and adjusted for the individual in the unit eyelane prior to NVGs operations. **(T-2)**. Do not wear NVGs during takeoff or landing. Do not don NVGs until at least 2000 feet AGL in climbing or level flight. In all cases, remove NVGs prior to the FAF.

3.20.3. NVGs equipped aircraft may operate below the ERAA/MSA down to a minimum of 1000 feet AGL during high-illumination (HI) periods (as defined by AFI 11-214). The flight lead or individual pilot is the final authority to assess actual illumination for a particular mission element, based on visibility and terrain features/resolution.

3.20.4. NVGs may be worn for night tanker rejoins, but will be raised to the up/stowed position or removed no later than the pre-contact position. **(T-2)**.

3.20.5. Unless required for battle damage checks or aircraft assistance, wingmen wearing NVGs will fly no closer than route formation. **(T-2)**.

3.20.6. Battle Damage Checks. NVGs will remain on. **(T-2)**. The aircraft performing the check will approach with position lights bright STEADY/FLASH or BEACONS on while the aircraft being checked sets external lights to a minimum, preferably OFF. **(T-2)**.

3.20.7. In-flight Emergencies with NVGs. During in-flight emergencies, immediately assess whether the NVGs aid or hinder completing emergency procedures. If they are a hindrance or the emergency may deteriorate into an ejection situation, remove and stow the NVGs.

3.20.8. Abnormal Procedures.

3.20.8.1. Lost sight**.** If sight loss occurs within a flight, execute appropriate lost wingman procedures. Consider highlighting position by increasing exterior lighting level, activating the afterburners, or deploying chaff/flares as the situation warrants.

3.20.8.2. NVGs failure. Ensure separation from other aircraft and the ground before attempting to remedy the NVGs failure.

3.20.8.2.1. Transition to instruments.

3.20.8.2.2. Perform lost wingman procedures if appropriate.

3.20.8.2.3. Route abort/climb above MSA if appropriate.

3.20.8.2.4. Terminate/KIO as applicable.

3.20.8.2.5. With other aircraft in the vicinity, direct them to raise their external lights to non-NVGs visible levels if two sources of deconfliction have not been established.

3.20.8.2.6. Attempt to regain NVGs operation by switching to the opposite battery. Once clear of other aircraft and terrain, change the battery. If these steps do not solve the problem, stow NVGs and proceed with non-NVGs plan.

3.20.8.3. Inadvertent flight into weather. Encountering poor weather conditions during NVG operations may cause loss of SA and aircrew distraction/disorientation.

3.20.8.4. Single ship or separated from flight members:

3.20.8.4.1. Transition to instruments.

3.20.8.4.2. Route abort if LOWAT, otherwise climb/descend to VMC.

3.20.8.4.3. Terminate/KIO as applicable.

3.20.8.5. Formation flight. If entering weather in formation/close proximity to other aircraft, perform the first five steps under NVGs failure, as appropriate, then climb/descend to attempt to regain VMC.

3.21. Fuel Requirements.

3.21.1. Joker Fuel. A pre-briefed fuel quantity needed to terminate an event and proceed with the remainder of the mission.

3.21.2. Bingo Fuel. A pre-briefed fuel quantity that allows the aircraft to return to the base of intended landing or alternate, if required, using preplanned recovery parameters and arriving with normal recovery fuel as listed below:

3.21.3. Normal Recovery Fuel. The fuel quantity on initial or at the FAF at the base of intended landing or alternate, if required. Fuel quantity will be as established locally or 2000 pounds, whichever is higher. **(T-2)**.

3.21.4. Declare the following when it becomes apparent that an aircraft may land at the intended destination or alternate, if required, with:

3.21.4.1. Minimum Fuel - 1200 pounds or less.

3.21.4.2. Emergency Fuel - 800 pounds or less.

3.22. Approaches and Landings.

3.22.1. The desired touchdown point for a VFR approach is 500 feet from the threshold or the glideslope interception point for a precision approach. When local procedures or unique conditions require landing beyond the normal touchdown point, adjust the touchdown point accordingly.

3.22.2. Minimum pattern and touchdown spacing between landing aircraft is 3000 feet for similar aircraft (e.g., F-15 versus F-15), 6,000 feet for dissimilar aircraft (e.g., F-15 versus F-16) or as directed by MAJCOM or the landing base, whichever is higher. Increase spacing to 6000 feet for similar and 9000 feet for dissimilar when wind conditions increase the likelihood of encountering wake turbulence or wake turbulence procedures are in effect.

3.22.3. Normally, all aircraft will land in the center of the runway and clear to the turnoff (cold) side of the runway when speed/conditions permit. **(T-2)**. Do not delay clearing to the cold side as failure to do so can create a conflict for subsequent landing aircraft.

3.22.4. Landing Restrictions.

3.22.4.1. When the computed landing roll exceeds 80 percent of the available runway, land at an alternate if possible.

3.22.4.2. Do not land over any raised web barrier (e.g., MA-1A, 61QSII).

3.22.4.3. During the aerobrake portion of a normal, dry runway landing, leave flaps down to provide increased aerodynamic drag and normal nose fall.

3.22.4.4. When the RCR at the base of intended landing is less than 12, land at an alternate if possible. If an alternate is not available, an approach end or midfield arrestment is recommended.

3.23. Overhead Traffic Patterns.

3.23.1. Overhead patterns can be made with unexpended practice ordnance and unexpended live air-to-air ordnance.

3.23.2. Initiate the break over the touchdown point or as directed.

3.23.3. Execute individual breaks in a level 180 degree turn to the downwind leg at minimum intervals of 5 seconds (except IP/SEFE chase or when in tactical formation).

3.23.4. Aircraft must be wings level on final at approximately 300 feet AGL and 1 mile from the planned touchdown point. **(T-2)**.

3.24. Tactical Overhead Traffic Patterns. Tactical entry to the overhead traffic pattern is permitted when:

3.24.1. The published overhead pattern altitude and airspeed are used.

3.24.2. Specific procedures are developed and coordinated with appropriate ATC agencies.

3.24.3. No more than four aircraft are in the flight. Aircraft/elements more than 6,000 feet in trail are considered a separate flight.

3.24.4. Wingmen in tactical formation offset the runway away from the direction of the break.

3.24.5. Normal downwind, base turn positions, and spacing are flown.

3.25. Touch-and-Go Landings.

3.25.1. Fly touch-and-go landings in accordance with AFTTP 3-3.F-15C, paragraph 10.17, as supplemented by MAJCOM. Touch-and-go landings are only practiced in F-15D aircraft, in accordance with AFI 11-202 Volume 3.

3.25.2. Do not fly touch-and-go landings with live or hung ordnance (except a safed 20mm gun) or with fuel remaining in any external tank.

3.26. Low Approaches. Observe the following minimum altitudes:

3.26.1. Normal/no flap single ship low approaches - so that touchdown does not occur.

3.26.2. IP/SEFE flying chase position - 50 feet AGL.

3.26.3. Formation low approaches and non-IP/SEFE chase - 100 feet AGL.

3.26.4. Chase aircraft during an emergency:

3.26.4.1. - 300 feet AGL unless safety or circumstances dictate otherwise.

3.26.4.2. During go-around, remain 500 feet below VFR overhead traffic pattern altitude until crossing the departure end of the runway unless local procedures, missed approach/climbout procedures, or controller instructions dictate otherwise.

3.27. Closed Traffic Patterns.

3.27.1. Initiate the pattern at the departure end of the runway unless directed/cleared otherwise by local procedures or the controlling agency.

3.27.2. When in formation, a sequential closed may be flown with ATC concurrence at an interval to ensure proper spacing.

3.27.3. Plan to arrive on downwind at 200-250 KIAS.

3.28. Back Seat Approaches and Landings.

3.28.1. A RCP landing qualified IP may accomplish back seat landings. An upgrading IP may only accomplish back seat landings when a RCP landing qualified IP is in the front cockpit.

3.28.2. During back seat approaches and landings, the front seat pilot will visually clear the area, monitor aircraft parameters and configurations, and be prepared to direct a go-around or take control of the aircraft (as briefed by the rear cockpit IP) if necessary. **(T-2)**.

3.29. Refer to Table 3.1. for a summary of weather minimums or pilot weather category (PWC) affecting F-15 operations.

Table 3.1. Weather Minimum Summary (Feet/NM/KM).

Event	Minimum
(USAFE) Formation Chase Approach	500 / 1.5NM / 2.4KM or PWC**
VFR Rejoin (Day)	1500 / 3NM / 4.8KM
VFR Rejoin (Night)	3000 / 5NM / 8KM
Low Level Navigation (Day)	1500 / 3NM / 4.8KM (notes 1&2)
Low Altitude Intercepts (Day)	3000 / 5NM / 8KM (note 3)
(USAFE) Touch-and-Go Landings	500/1.5NM / 2.4KM
(USAFE) Approach to Field w/o Approved DoD Minimums	Must remain VMC throughout entire Approach (note 4)
** Whichever is higher	
Notes: 1. Unless national rules are higher. 2. 2000/8 in countries where the minimum altitude is 1000 feet AGL. 3. 3500/8 in countries where minimum altitude is 1000 feet AGL. 4. Does not apply if USAFE/A3 waiver is approved for approach being flown.	

3.30. Refer to Table 3.2. for a summary of minimum altitudes that affect F-15 operations.

Table 3.2. Minimum Altitude Summary (in Feet AGL).

Event	Minimum
Aerobatics/ACBT/Stalls	5000
Lead Change	See note
Chase (Emergency)	300
Chase (FE/IP)	50
Formation Low Approach	100
Low Approaches	So as to not touchdown
KIO	1000
Fuel Dumping (non in-flight emergency (IFE))	10000
NOTE: 1000 feet over water; 1500 feet at night /IMC unless on radar downwind	

Chapter 4

INSTRUMENT PROCEDURES

4.1. Approach Category.

4.1.1. The F-15 is approach Category E. Accomplish missed approach in accordance with flight manual procedures. Missed approach airspeed is 200-250 KIAS.

4.1.2. Use approach Category D minimums at an emergency/divert airfield where no Category E minimums are published. Airfields with Category D minimums may be designated as an alternate (divert) airfield. Further, practice instrument approaches may be flown using Category D minimums if VMC can be maintained throughout the procedure. Under these circumstances, approach Category D minimums may be used provided:

4.1.2.1. A straight-in approach is flown.

4.1.2.2. The aircraft is flown at final approach airspeed of 165 KIAS or less.

4.1.2.3. The aircraft is flown at 255 knots true airspeed (KTAS) or less for the missed approach segment of the approach. At high-pressure altitudes and temperatures, 255 KTAS may not be compatible with published missed approach airspeeds and Category D approaches should not be flown.

4.1.2.3.1. **(USAFE/AFAFRICA)** F-15C/D aircraft may utilize approach Category D minimums when conducting USAFE/AFAFRICA tasked tempo temporary duty (TDY)s or Flying Training Deployments (FTDs) to airfields without published Category E minimums, with prior OG/CC or equivalent authority notification in order to accomplish the USAFE/AFAFRICA mission more efficiently.

4.1.2.4. The aircraft has F-15 navigational equipment capabilities.

4.1.2.5. When filing a DD Form 175 *Military Flight Plan*, use type designator/suffix codes which denote the transponder/navigational aid capabilities (in accordance with FLIP general planning).

4.1.2.6. When filing a DD Form 1801 (DoD International Flight Plan), use Equipment and Capabilities section with applicable abbreviations (in accordance with FLIP general planning).

4.1.2.7. F-15 with Embedded global positioning system/inertial navigation system (EGI):

4.1.2.7.1. Type designator (TD)/suffix code: F-15/G (applicable to DD Form 175).

4.1.2.7.2. Equipment and Capabilities: D G I L T U / C (applicable to DD Form 1801).

4.1.2.7.3. Not reduced vertical separation minimum (RVSM) equipped; however, allowed to operate in RVSM airspace with ATC approval.

4.1.2.7.4. Approved for Enroute Area Navigation (RNAV). May file RNAV routings using point-to-point routing, T-Routes, or Q-Routes. Filing any RNAV routing requires use of the DD Form 1801.

4.1.2.7.5. Not approved for RNAV approaches.

4.1.2.8. F-15 without EGI:

4.1.2.8.1. TD/suffix code: F-15/I (applicable to DD Form 175).

4.1.2.8.2. Equipment and Capabilities: D I L T U / C (applicable to DD Form 1801).

4.1.2.8.3. Not RVSM equipped; however, allowed to operate in RVSM airspace with ATC approval.

4.1.2.8.4. Approved for RNAV. Can only file RNAV routings using point-to-point routing. Cannot file T-Routes or Q-Routes which require Global Navigation Satellite System (GNSS) capabilities in accordance with IFR enroute Low/high altitude charts. Filing any RNAV routing requires use of the DD Form 1801.

4.1.2.8.5. Not approved for RNAV approaches.

4.1.3. Due to a variety of host nation and International Civil Aviation Organization (ICAO) procedures which may differ from standard F-15 procedures, pilots must ensure that the airspeeds they plan to fly on instrument approach procedures are compatible with both ICAO/terminal instrument procedures (TERPS) and safety of flight requirements. **(T-0)**.

4.2. Takeoff and Join-up.

4.2.1. The flight leader must get an appropriate ATC clearance (altitude block or trail formation) when a flight join-up is not possible due to weather conditions or operational requirements. **(T-2)**. Formation trail departures must comply with instructions for a nonstandard formation flight as defined in FLIP. **(T-2)**.

4.2.2. If weather is below 1500 feet and 3 miles, each aircraft/element will climb on takeoff heading to 1000 feet AGL before initiating any turns, except when departure instructions specifically preclude compliance. **(T-2)**.

4.3. Trail Procedures.

4.3.1. **General.**

4.3.1.1. During trail formations, basic instrument flying is the first priority and must not be sacrificed when performing secondary trail tasks. **(T-2)**. Strictly adhere to the briefed airspeeds, power settings, altitudes, headings, and turn points. If task saturation occurs, cease attempts to maintain trail, concentrate on flying the instrument departure, and then notify the flight lead. The flight lead will then notify ATC. **(T-2)**.

4.3.1.2. Flight leaders will request non-standard formation from ATC. **(T-2)**.

4.3.1.3. ATC instructions issued to the lead aircraft apply to the entire flight.

4.3.1.4. Flight leads will brief aircraft/element spacing. **(T-2)**. Minimum spacing between aircraft when in non-standard formation is 9,000 feet and will be maintained using on-board radar systems. **(T-2)**.

4.3.1.5. Each aircraft/element will follow the No Radar Contact procedures until the aircraft/element immediately in trail has radar contact and called “tied.” **(T-2)**.

4.3.2. No Radar Contact.

4.3.2.1. The flight leader will call initiating all turns. **(T-2)**. Subsequent aircraft must delay turns to maintain the desired spacing. **(T-2)**.

4.3.2.2. Each aircraft/element will maintain 20 seconds or 2-3 mile spacing using all available aircraft systems and navigational aids to monitor position. **(T-2)**.

4.3.2.3. During climbs and descents, each aircraft/element will call passing each 5000 foot altitude increment with altitude and heading (or heading passing) until join-up, level-off, or the following aircraft/element calls "tied." **(T-2)**.

4.3.2.4. Each aircraft/element will call initiating any altitude or heading change. **(T-2)**. Acknowledgments are not required; however, it is imperative that preceding aircraft/elements monitor the radio transmissions and progress of the succeeding aircraft/elements and immediately correct deviations from the planned route.

4.3.2.5. Each aircraft/element will maintain at least 1000 feet vertical separation from the preceding aircraft/element until establishing radar/visual contact, except in instances where departure instructions specifically preclude compliance. **(T-2)**. Reduce vertical separation to 500 feet if necessary to comply with MSA restrictions.

4.3.2.6. In the event a visual join-up cannot be accomplished on top or at level off, the flight leader will request altitude separation for each succeeding aircraft/element to meet the requirements of the **paragraph 4.3.2.5** **(T-2)**.

4.3.3. Radar Contact.

4.3.3.1. Each aircraft/element will call "tied" when radar contact is established with the preceding aircraft. **(T-2)**. Once all aircraft are tied, no further radio calls are required, unless radar contact is lost. If ATC directs a question, or guidance to a certain aircraft, that pilot may respond directly to ATC. Flight leads will ensure that wingmen are aware of any unplanned or ATC directed intermediate level off altitudes. **(T-2)**.

4.3.3.2. In flights of three or more aircraft, use all available aircraft systems (e.g., radar, TACAN, AAI, FDL, etc.) to ensure that trail is maintained on the correct aircraft.

4.3.4. Trail Departures.

4.3.4.1. Use a minimum of 20 seconds takeoff spacing. **(T-2)**.

4.3.4.2. Each aircraft/element will accelerate in MIL/AB power until reaching 350 KIAS. **(T-2)**. Climb at 350 KIAS until reaching cruise mach/KTAS, unless otherwise briefed.

4.3.4.3. Upon reaching 350 KIAS, the flight leader will set 850 FTIT unless otherwise briefed. **(T-2)**.

4.3.4.4. Limit all turns to a maximum of 30 degrees of bank.

4.3.5. Enroute Trail**.** Flight leads must brief airspeeds, power settings, and configurations. **(T-2)**.

4.3.6. Trail Recovery.

4.3.6.1. Trail recovery procedures must be coordinated/approved through the responsible ATC facilities and addressed in a local operating procedure or in the unit supplement to this volume. **(T-2)**. Trail recoveries will only be accomplished at home stations/deployed locations where procedures have been established and briefed. **(T-2)**. As a minimum, procedures must address each recovery profile, missed approach, climbout, lost contact, lost communications and desired/maximum spacing requirements. **(T-2)**.

4.3.6.2. Limit trail recovery to a maximum of four aircraft.

4.3.6.3. Trail recoveries are authorized when weather at the base of intended landing is at/above the highest pilot weather category in the flight or approach minimums, whichever is higher.

4.3.6.4. Trail recoveries will not terminate in simultaneous precision approach radar (PAR) or Airport Surveillance Radar (ASR) approaches. **(T-2)**. Recoveries to separate PAR/ASRs are authorized, flights will split prior to PAR/ASR final. **(T-2)**.

4.3.6.5. The flight lead must brief the flight on spacing, configuration, and airspeeds. **(T-3)**. Minimum spacing between aircraft is 9,000 feet in IMC and will be maintained using on-board radar. **(T-2)**.

4.3.6.6. The flight lead must coordinate the trail recovery with ATC prior to taking spacing. **(T-2)**.

4.3.6.7. Prior to split-up, the flight lead must ensure that all wingmen have operative navigational aids and air-to-air radar. **(T-2)**.

4.3.6.8. Formation break-up will be accomplished in accordance with **paragraph 4.4** **(T-2)**. Flight separation will be in accordance with local ATC directives. **(T-2)**.

4.3.6.9. The formation must squawk as directed by ATC. **(T-2)**.

4.3.6.10. ATC instructions to the lead aircraft will be for the entire flight. **(T-2)**. ATC will provide radar flight following for the entire formation. **(T-2)**.

4.3.6.11. Limit all turns to a maximum of 30 degrees of bank.

4.3.6.12. Once established on a segment of a published approach, each aircraft must comply with all published altitudes and restrictions while maintaining in-trail separation. **(T-0)**.

4.3.6.13. Unless local procedures establish defined reference points for airspeed/ configuration changes, the flight lead must direct changes by radio. **(T-2)**. At flight lead's call, all aircraft must simultaneously comply with the directed change. **(T-2)**.

4.3.6.14. All aircraft must report the FAF. **(T-2)**.

4.3.6.15. If contact is lost with the preceding aircraft, the pilot will transmit "Call Sign (C/S)) lost contact." **(T-2)**. The preceding aircraft will respond with altitude, airspeed and heading. **(T-2)**. Establish altitude deconfliction and coordinate a separate clearance with ATC. If contact is lost while established on a segment of a published approach, flight members may continue the approach, but must confirm separation via navigation

aids. **(T-0)**. If separation cannot be confirmed, execute missed approach or climbout as instructed by ATC.

4.4. Formation Break-up.

4.4.1. If possible, accomplish formation break-up in VMC. If IMC, accomplish the break-up in straight and level flight. Prior to a break-up in IMC, the flight leader must transmit attitude, airspeed, altitude, and altimeter setting that will be acknowledged by wingmen. **(T-2)**.

4.4.2. Wingmen must confirm good navigational aids. **(T-2)**.

4.5. Formation Penetration.

4.5.1. Restrict formation penetrations in route/close formation to two aircraft when the weather at the base of intended landing is less than overhead traffic pattern minimums.

4.5.2. Formation penetrations using radar trail procedures are authorized when weather at the base of intended landing is at/above the highest pilot weather category in the flight or approach minimums, whichever is higher.

4.6. Formation Approach. During IMC formation flights, do not change lead or wing positions below 1500 feet AGL unless on radar downwind.

4.7. Simulated Instrument Flight. Simulated instrument flight requires a qualified safety observer in either cockpit of the aircraft or in a chase aircraft. Use all means to clear the area for hazards.

4.7.1. A safety observer in the same aircraft must have an operable intercom. **(T-2)**.

4.7.2. A chase aircraft is required in order to log simulated instrument flight in an F- 15A/C. This does not preclude flying approaches in VMC without a chase, however, in this case place primary emphasis on the "See and Avoid" concept. Chase aircraft will be in a position where they can effectively clear and/or provide assistance. **(T-2)**.

4.8. Use of the Heads Up Display (HUD). For the F-15A/B/C/D, use the HUD as an additional instrument reference not as the sole instrument reference. Do not use the HUD to recover from an unusual attitude except when no other reference is available.

4.9. Use of the Joint Helmet Mounted Cueing System (JHMCS). Although JHMCS displays sufficient aircraft flight symbology to provide basic situational awareness, it is not intended to be used for primary flight instrumentation. Do not use JHMCS to recover from an unusual attitude.

Chapter 5

AIR-TO-AIR WEAPONS EMPLOYMENT

5.1. References. AFI 11-214 contains air-to-air procedures, to include operations with live ordnance applicable to all aircraft. This chapter specifies procedures or restrictions applicable to F-15 operations.

5.2. Simulated Gun Employment. Missions may be flown with a loaded gun provided the gun is safe in accordance with Technical Order (T.O.) 1F-15C-34-1-3CL-1, *Nonnuclear Weapon Delivery Checklist 1*, and a trigger check is first performed with the master arm switch in arm with Training Mode deselected. Point the aircraft away from other aircraft and inhabited areas during the trigger check. Do not perform a trigger check with a hot gun. If an aircraft is flown with a hot gun or live missiles, the procedures in AFI 11-214 apply.

5.3. Maneuvering Limitations.

5.3.1. Minimum airspeed during low altitude offensive or defensive maneuvering (LOWAT) is 350 KIAS.

5.3.2. When configured with three external fuel tanks, aircraft will operate under the limited maneuvering category as defined in AFI 11-214. **(T-2)**. When configured with external wing tanks and no centerline tank, unlimited maneuvering is allowed once the externals are empty (flight manual restrictions still apply). The tank restriction does not apply to aerial gunnery tow aircraft. Tow aircraft may fly using any approved (for test and evaluation) or certified (operational) configurations.

5.3.3. Negative-G gun jinks are prohibited.

5.3.4. When acting as a restricted maneuvering target for low altitude intercepts, the minimum airspeed is 220 KIAS or 20 units AOA, whichever produces the higher airspeed.

5.4. Aerial Gunnery Tow Procedures.

5.4.1. AFI 11-214 applies. In addition:

5.4.1.1. Aerial Gunnery Target Set (AGTS)-36 Target Set Deployment. Deploy the target set over areas clear of surface activity, near the range where shooting will occur. **(T-2)**. The tow will maintain 240 +/- 10 KIAS and ensure other flight members are clear during deployment. **(T-2)**. A chase aircraft will ensure that the AGTS-36 tow reel turbine doors open for deployment and the visual augmenter (VA) deploys properly. **(T-2)**. Any time the target set becomes unstable or flies erratic, the chase aircraft will advise the tow to cut the target loose. **(T-2)**.

5.4.1.2. AGTS-36 Target Set Reel-In. The tow will slow to 240 +/- 10 KIAS and initiate target set reel-in when shooting is complete. **(T-2)**. A chase will evaluate target set stability and advise the tow of any abnormalities. **(T-2)**. Recover target sets that have been hit if they do not exhibit abnormal flight characteristics. However, if a target set becomes unstable or flies erratic during reel-in, it should be cut loose. Initiate reel-in over areas clear of surface activity in anticipation of VA release. The chase will ensure the tow reel turbine doors are closed, the target set locks into place, the VA releases and perform a battle damage assessment (BDA) on the tow after reel-in is complete. **(T-2)**.

5.4.1.3. Target Designation Unit (TDU)-32 A/B Banner Deployment. The banner departs the airfield fully deployed. Enroute to the range, chase aircraft will ensure a stable flying banner. **(T-2)**. While in straight and level flight, the chase pilot will also pass the direction of "banner lean" in clock position to the tow pilot. **(T-2)**. This will establish the best turn direction for employment. (If banner is flying at the 10 o'clock position; tow will turn right.) **(T-2)**.

5.4.1.4. TDU-32 A/B Banner Recovery. The tow will return to the drop zone in accordance with locally approved procedures for banner release. **(T-2)**. Tower will clear the tow to the "drop zone" and the pilot will release the banner for recovery by the ground crew. **(T-2)**.

5.4.1.5. Abnormal Procedures.

5.4.1.5.1. VA fails to deploy. Reel the target set back in and do not use for firing. **(T-2)**.

5.4.1.5.2. VA fails to release after reel-in. The tow must recover using hung ordnance procedures. **(T-2)**.

5.4.1.5.3. Erratic or unstable target set. In all cases where target sets exhibit unstable or erratic flight characteristics, the target set should be cut loose. Avoid trying to reel-in a target set that begins to become unstable/erratic, cut it free with whatever length of cable is present at the time it begins to fly abnormally.

5.4.1.5.4. Target set shot off. Cut the remaining cable off. The cable may jam in the tow reel if reel-in is attempted without a target set present.

5.4.1.5.5. Cutting target sets loose. Use primary and secondary cutters as the preferred means to cut target sets loose. If this is unsuccessful, reel out the target set until the cable/target set fall off. The next options depend on fuel availability, weather and resources. The tow may try to burn the cable off using afterburner and high AOA maneuvers. The target set may be dragged off in the water. In this case, the chase will fly no lower than 1000 feet AGL and the tow no lower than 400 feet AGL. **(T-2)**. If all other means of cutting the target set loose are unsuccessful and the target set may not be safely recovered, jettison the AGTS-36 tow reel.

5.4.1.5.6. Landing with cable remaining (no target set). The tow must recover using hung ordnance procedures. **(T-2)**. Plan the landing to avoid damage to runway approach lighting.

5.4.1.5.7. Controlled Jettison of Banner on Takeoff. If circumstances permit, tow aircraft jettisons banner in accordance with local jettison procedures.

5.4.1.5.8. Unable to Release Banner**.** If possible, gain chase aircraft (shooter desired) and proceed to local jettison area and attempt jettison. If banner still remains, recover in accordance with local procedures or use AGTS cable remaining procedures.

5.4.1.5.9. Erratic/Unrecoverable Banner. Jettison in working area.

5.4.1.5.10. Banner Shot Off, Cable Remaining. Release cable in working area. Shooter rejoins with tow to ensure no cable remains. If cable remains, follow procedures in **paragraph 5.4.1.5.6** NORAD VID Procedures. **(T-2)**.

5.4.2. Turn the HUD camera/digital video recorder (DVR) on during active air defense visual identification passes no later than 1 NM in the target's stern and leave it on until after completing the breakaway maneuver.

Chapter 6

AIR-TO-SURFACE WEAPONS EMPLOYMENT

6.1. General.

6.1.1. AFI 11-214, *Air Operations Rules and Procedures*, contains air-to-surface procedures and restrictions applicable to all aircraft. This chapter contains procedures and restrictions specific to F-15 operations.

6.1.2. Air-to-Surface weapons employment includes both live and simulated attacks during high angle strafe (HAS), basic surface attack (BSA) and maritime air support (MAS) missions including A/G weapons employment against maritime targets.

6.2. Weapons Delivery Parameters. The following parameters and requirements form the basic framework for weapons delivery training and all deliveries will conform to limits established for each specific event. **(T-2)**.

6.2.1. Gunnery Events. Procedures, training rules, and foul criteria are contained in AFI 11-214. For over-land training, only preplanned targets will be attacked to ensure target elevation is known and safe escape altitudes are valid. **(T-2)**.

6.2.1.1. High Angle Strafe.

6.2.1.2. Minimum recovery altitude is at or above pilot LOWAT certified category minimums (day).

6.2.1.3. Minimum recovery altitudes are in accordance with **AFI 11-214 Chapter 5** (night).

6.3. Simulated Attacks against Range, Off-Range or Manned Targets:

6.3.1. May be conducted, to include use of master arm when carrying practice ordnance under the following restrictions:

6.3.2. No live A/A missiles are loaded.

6.3.3. Simulated Strafe.

6.3.4. Simulated strafe is defined as the combined use of master arm and trigger actuation during a practice gun attack against ground targets.

6.3.4.1. Simulated strafe is permitted with a safed gun, loaded or unloaded, provided a trigger check is previously accomplished.

6.3.4.2. Do not perform simulated strafe with a hot gun (loaded or empty).

Chapter 7

ABNORMAL OPERATING PROCEDURES

7.1. General. Follow the procedures in this chapter when other than normal circumstances occur. These procedures do not supersede flight manual guidance.

7.1.1. Do not accept an aircraft for flight with any problem that could jeopardize safe flight.

7.1.2. Do not use a malfunctioning system unless it is required for safe recovery of the aircraft. Do not continue in-flight troubleshooting a malfunction after completing flight manual emergency procedures and the aircraft may be safely recovered.

7.1.3. Only conduct fuel dumping to reduce aircraft gross weight for safety of flight. When circumstances permit, dump above 10000 feet AGL over unpopulated areas. Ensure the dump switch is returned to normal before landing. Make the appropriate entry in the AFTO Form 781A, *Maintenance Discrepancy and Work Document.*

7.1.4. Do not taxi aircraft with malfunctions that affect the nose wheel steering or brake system.

7.1.5. For actual or perceived flight control malfunctions, terminate maneuvering and take appropriate action. If the flight control problem was due to crew member/passenger stick or rudder interference in B/D model aircraft, the pilot will take appropriate action to ensure no further flight control interference occurs. **(T-2)**.

7.2. Ground Aborts.

7.2.1. If a flight member aborts prior to takeoff, the flight leader will normally renumber the flight. Flight leaders must advise the appropriate agencies of such changes. **(T-2)**.

7.2.2. In the event of an abort, formation flight may only continue if it is led by a qualified flight lead. The alternatives are a sympathetic abort or proceeding on a pre-briefed single-ship mission.

7.2.3. Delayed aircraft may join the flight at a briefed rendezvous point or may fly a briefed alternate single ship mission. If accomplishing a join-up, cease tactical maneuvering until the delayed aircraft is joined and all flight members are ready to continue. **(T-2)**.

7.3. Takeoff Aborts.

7.3.1. If aborting the takeoff, clear to the appropriate side of the runway as expeditiously as possible based on position within the element. **(T-2).** If this is not feasible because of possible cable engagement, clear straight-ahead. **(T-2).** As soon as practicable, give call sign and state intentions. **(T-2)**. Call "Cable, Cable, Cable" to indicate a departure-end arrestment. **(T-2)**. Following aircraft hold their position, abort or takeoff as appropriate to maintain adequate clearance. **(T-2)**.

7.3.2. In accordance with T.O. 1F-15A-1, *Preliminary Flight Manual*, consideration should be given to hot brakes when brakes are applied above 100 KIAS. Ground speed, runway length, gross-weight, runway condition, heavy braking and outside air temperature are all variable factors which could contribute to hot brakes. If hot brakes are suspected; declare a

ground emergency, taxi the aircraft to the designated hot brake area, and follow hot brake procedures.

7.3.3. If aborting a takeoff lower the tailhook if there is any doubt about the ability to stop on the runway.

7.3.4. Following an aborted takeoff, consider having ground maintenance personnel check the aircraft for hot brakes.

7.4. Air Aborts.

7.4.1. If an abort occurs after takeoff, all aircraft will maintain their original numerical call sign. **(T-2).**

7.4.2. Escort aborting aircraft with an emergency to the field of intended landing. In other cases, the flight leader will determine if an escort is required. **(T-2)**.

7.4.3. Abort the mission and land out of a straight-in approach, regardless of apparent damage or subsequent normal operation, for any of the following:

7.4.3.1. Birdstrike/foreign object damage.

7.4.3.2. Flight control system anomalies. This does not include flight control system lights that reset in accordance with flight manual procedures.

7.4.3.3. Engine flameout, stagnation or shutdown.

7.4.4. If an aircraft experiences an over-G, use the following procedures:

7.4.4.1. Perform a battle damage check after all over-Gs.

7.4.4.2. Non-OWS equipped aircraft and OWS equipped aircraft whose OWS is not operational.

7.4.4.2.1. Terminate the mission and land as soon as practicable from a straight-in approach.

7.4.4.3. Aircraft equipped with an operable OWS system that experiences an over-G.

7.4.4.3.1. Immediately terminate maneuvering and call up the OWS matrix on the VSD to analyze the displayed parameters.

7.4.4.3.2. If a positive G, level “1” (one) is displayed in any column of the matrix, except mass items (MIT), perform a battle damage check with emphasis on the overall condition of the aircraft. If no abnormalities are noted, the flight lead may continue the briefed mission. If a subsequent positive G, level “1” or greater over-G occurs, terminate the mission, perform a battle damage check, RTB and fly a straight-in approach.

7.4.4.3.3. If a positive G, level “2” (two) or greater; or a negative G, any level, is displayed in any column of the matrix, terminate the mission, perform a battle damage check, RTB and fly a straight-in approach.

7.4.4.4. Document actual over-Gs (positive G, level 1 or greater, or negative G, any level) in the AFTO Form 781A, *Maintenance Discrepancy and Work Document* after flight.

7.4.5. In the F-15B/D, the pilot in command is primarily responsible for handling in-flight emergencies. The additional aircrew member will provide checklist assistance at the request of the pilot in command. **(T-2)**.

7.5. Engine Malfunctions. Report all engine anomalies during maintenance debriefing. **(T-2)**.

7.6. Radio Failure.

7.6.1. **General.** Individual aircraft experiencing radio failure will comply with procedures outlined in FLIP, AFI 11-202 Volume 3, this instruction, and local directives. **(T-2)**.

7.6.2. **Formation:**

7.6.2.1. Flight members who experience total radio failure while in close or route formation will attract the attention of another flight member and give the appropriate visual signals. **(T-2)**. Terminate the mission as soon as practicable and lead the NORDO aircraft to the base of intended landing or a divert base. Perform a formation approach to a drop-off on final unless safety considerations dictate otherwise.

7.6.2.2. If flying other than close/route formation when radio failure occurs, the NORDO aircraft should attempt to rejoin to a route position at approximately 500 feet on another flight member. The NORDO aircraft is responsible for maintaining clearances from other flight members until her or his presence is acknowledged by a wingrock, signifying clearance to join. Once joined, the NORDO aircraft will give the appropriate visual signals. If pre-briefed, the NORDO aircraft may proceed to a rendezvous point and hold. If no one has rejoined prior to reaching bingo fuel, the NORDO aircraft should proceed to the base of intended landing or a divert base. Aircraft experiencing any difficulty/emergency in addition to NORDO will proceed as required by the situation. **(T-2)**.

7.6.3. **Aerial Gunnery/Air-to-Surface Gunnery/Missile Firing NORDO Procedures:**

7.6.3.1. Aircraft will not fire without two-way radio contact. **(T-2)**.

7.6.3.2. Shooting aircraft: safe the armament switches, join on another member of the flight or the tow aircraft, in accordance with **paragraph 7.6.2**

7.6.3.3. Aerial gunnery tow aircraft: rocks wings and continues the turn if an attack is in progress. The flight leader of the attacking aircraft will "knock off" the attack and join on the tow's wing, remaining clear of the target in the event it is cut. **(T-2)**. The tow pilot will use standard hand signals to indicate her or his difficulty. **(T-2)**. The flight leader will signal when the target is cleared for cut with a slicing motion across the throat. **(T-2)**. After the target is away and the flight lead determines there is no remaining cable, the tow will RTB with an escort following the briefed NORDO recovery procedures. **(T-2)**. If cable remains, follow local procedures.

7.6.4. NORDO Recovery:

7.6.4.1. Apply the procedures in AFPAM 11-205 and FLIP.

7.6.4.2. If a formation straight-in approach is flown and a go-around becomes necessary, the chase will go-around, pass the NORDO aircraft and rock her or his wings. **(T-2)**. The NORDO aircraft will go-around, if the situation allows. **(T-2)**. If the NORDO

aircraft is in formation as a wingman, the leader will initiate a gentle turn into the wingman and begin the go-around. **(T-2)**.

7.6.4.3. A NORDO aircraft intending to make an approach-end cable engagement should signal the escorting aircraft by extending the tailhook. If the NORDO aircraft is not escorted, the pilot should fly a straight-in approach flashing the landing light on final to signal the tower.

7.7. Severe Weather Penetration. Avoid flight through severe weather. If unavoidable, flights should break-up and obtain separate clearances prior to severe weather penetration.

7.8. Lost Wingman Procedures. In any lost wingman situation, immediate separation of aircraft is essential. Upon losing sight of the leader or if unable to maintain formation, the wingman will simultaneously:

7.8.1. Execute the applicable lost wingman procedures. Refer to **paragraph 7.9** for specific spatial disorientation (SD) considerations. Smooth application of control inputs is imperative to minimize the effects of SD.

7.8.2. Transition to primary flight instruments. **(T-2)**.

7.8.3. Inform lead by transmitting "C/S is lost wingman." **(T-2)**.

7.8.4. After executing a lost wingman procedure, do not attempt rejoining with the flight until obtaining permission from the flight lead. **(T-2)**.

7.8.5. When able, obtain a separate clearance. **(T-2)**.

7.8.6. Observe all published terrain clearance limits. **(T-2)**.

7.8.7. Two-Ship or Three-Ship Flights (three-ship echelon, refer to four-ship procedures):

7.8.7.1. Wings-level Flight (climbing, descending, or straight and level)**.** Turn away using 15 degrees of bank for 15 seconds, then resume original heading.

7.8.7.2. Turns:

7.8.7.2.1. Outside the Turn. Reverse the direction of turn using 15 degrees of bank for 15 seconds. Continue straight ahead to ensure separation prior to resuming the turn.

7.8.7.2.2. Inside the Turn. Momentarily reduce power to ensure nose-tail separation and direct the flight leader to roll out of the turn. Maintain the original turn. The leader may only resume the turn when separation is ensured.

7.8.7.2.3. Final Approach. Momentarily turn away from lead to ensure clearance and execute the published missed approach procedure.

7.8.7.2.4. Missed Approach. Momentarily turn away from lead to ensure clearance and continue the published or assigned missed approach procedure. Climb to 500 feet above missed approach altitude.

7.8.7.3. Four-Ship Flights. Number 2 and 3 follow the procedures outlined above. Number 4's initial action assumes that number 3 has also gone lost wingman. In addition to **paragraph 7.8** number 4 will:

7.8.7.3.1. Wings-Level Flight (climbing, descending, or straight and level). Turn away using 30 degrees of bank for 30 seconds, then resume the original heading. **(T-2)**.

7.8.7.3.2. Turns:

7.8.7.3.2.1. Outside the Turn. Reverse direction of turn using 30 degrees of bank for 30 seconds to ensure separation from lead and number 3.

7.8.7.3.2.2. Inside the Turn. Momentarily reduce power to ensure nose-tail separation and increase bank angle by 15 degrees. Direct the leader to roll out. The leader will only resume the turn when separation is ensured. **(T-2)**.

7.8.7.4. Leader must acknowledge the lost wingman's radio call and, when appropriate, transmit attitude, heading, altitude, airspeed, and other parameters. **(T-2)**.

7.8.7.5. If a wingman becomes separated and any aircraft experiences radio failure, the aircraft with the operational radio will obtain a separate clearance. **(T-2)**. The NORDO aircraft will turn the IFF/SIF to NORMAL code 7600 while proceeding with previous clearance. **(T-2)**. If an emergency situation arises along with radio failure, turn the IFF/SIF to EMERGENCY for the remainder of the flight.

7.8.7.6. Only practice lost wingman procedures in VMC.

7.9. Spatial Disorientation (SD). Conditions that prevent a clear visual horizon or increase pilot tasking are conducive to SD. To prevent SD, the pilot must increase her or his instrument crosscheck rate. (T-2). If SD symptoms are encountered:

7.9.1. Single Ship:

7.9.1.1. Concentrate on flying basic instruments with frequent reference to the attitude indicator. Use heads-down instruments. If flying dual, consider transferring control to the other crewmember.

7.9.1.2. If symptoms persist and conditions permit, fly straight and level flight until symptoms abate, usually within 60 seconds. Consider using the autopilot.

7.9.1.3. If necessary, declare an emergency and advise ATC.

7.9.1.4. **NOTE:** It is possible for SD to proceed to the point where the aircrew is unable to see or interpret the flight instruments. In this situation, aircraft control may be impossible. If this occurs, the aircrew should consider ejecting.

7.9.2. Formation Lead:

7.9.2.1. Advise the wingmen that he has SD and comply with procedures in **paragraph 7.9.1.1**

7.9.2.2. Use the wingmen to confirm attitude and provide verbal feedback.

7.9.2.3. If symptoms persist, terminate the mission and recover the flight by the simplest and safest means possible.

7.9.3. Formation Wingman:

7.9.3.1. Advise lead of the disorientation.

7.9.3.2. Lead will advise wingman of aircraft attitude, altitude, heading, and airspeed. **(T-2)**.

7.9.3.3. If symptoms persist and conditions permit, lead will establish straight and level flight for 30-60 seconds. **(T-2)**.

7.9.3.4. If the above procedures are not effective, lead should consider passing the lead to the wingman, provided the leader will be able to maintain situational awareness from a chase position. **(T-2)**. Transfer lead while in straight and level flight. Once assuming the lead, maintain straight and level flight for 60 seconds. If necessary, terminate the tactical mission and recover by the simplest and safest means possible.

7.9.4. Greater than 2-Ship Formation. Lead should separate the flight into elements to more effectively handle a wingman with persistent SD symptoms. Establish straight and level flight in accordance with **paragraph 4.4** (Formation Break-up). Plan to keep the element with the SD pilot straight and level while the other element separates.

7.10. Armament System Malfunctions.

7.10.1. Inadvertent Release.

7.10.1.1. Record switch positions at the time of inadvertent release and provide to armament and safety personnel. Record the impact point, if known.

7.10.1.2. Safe the armament switches and do not attempt further release in any mode. Treat remaining stores as hung ordnance and follow hung ordnance procedures during RTB.

7.10.1.3. If remaining stores present a recovery hazard, jettison them in a suitable area on a single pass, if practicable.

7.10.2. Failure to Release/Hung Ordnance. If ordnance fails to release when all appropriate switches are set, proceed as follows.

7.10.2.1. Hung live ordnance or aircraft malfunction that precludes further live weapons delivery.

7.10.2.1.1. Attempt to release store(s) using an alternate delivery mode. (Not Applicable (N/A) for air-to-air WSEP.) If unsuccessful, attempt to jettison store(s) using selective jettison procedures. Lastly, consider attempting to selectively jettison the pylon if ordnance is unsecured or security cannot be determined.

7.10.2.1.2. If ordnance remains on the aircraft, follow the hung ordnance recovery procedures in **paragraph 7.10.4**

7.10.3. Hangfire/Misfire:

7.10.3.1. A missile that fires but fails to depart the aircraft is a hangfire. If this occurs, the missile should be closely observed and safety checked by a chase pilot.

7.10.3.2. A missile that fails to fire when all appropriate switches were selected is a misfire. If this occurs, safe the master arm switch and follow the hung ordnance recovery procedures.

7.10.4. Hung Ordnance/Weapons Malfunction Recovery:

7.10.4.1. If practicable, visually inspect the aircraft for damage.

7.10.4.2. Declare an emergency for hung live ordnance (not required for hung practice/inert ordnance or for live unexpended ordnance).

7.10.4.3. If available, obtain a chase aircraft (N/A at night) and avoid populated areas and trail formations.

7.10.4.4. Land from a straight-in approach.

7.10.5. Miscellaneous Procedures:

7.10.5.1. Pilots will not attempt to expend ordnance using a delivery system with a known weapons release malfunction. **(T-2)**.

7.10.5.2. When abnormal missile launch or erratic missile flight is noted after launch, visually inspect the launching aircraft by another pilot (if possible) to determine if any damage has occurred.

7.11. Post Arresting Gear Engagement Procedures.

7.11.1. Do not shut down the engine(s) unless directed by the ground crew, there is a fire, or other conditions dictate. **(T-2)**.

7.11.2. Raise the tailhook on the ground crew's signal.

7.11.3. Do not taxi until directed.

7.11.4. Comply with local directives.

7.12. In-flight Practice of Emergency Procedures.

7.12.1. Simulated emergency procedure is defined as any procedure that produces an effect that closely parallels an actual emergency, such as retarding the throttle to simulate the drag equivalent to a flamed out engine.

7.12.2. Aborted Takeoff Practice. Only practice aborted takeoffs in the flight simulator, cockpit procedures trainer (CPT), or, if the trainer is unavailable, a static aircraft.

7.12.3. Simulated in-flight loss of both engines is prohibited.

7.12.4. Practice in-flight engine shutdown is prohibited.

7.12.5. Emergency Landing Patterns (Refer to AFI 11-202 Volume 3).

7.12.5.1. Field Requirements. Practice of emergency landing patterns at active airfields is authorized provided that crash rescue and ATC facilities are available and in operation.

7.12.5.2. Supervisory Requirements**.** IQT (including Formal Training Unit (FTU)) pilots require an IP on board the aircraft or in a chase aircraft. MQT (including FTU) pilots require a SOF in place and an IP or flight lead monitoring from the traffic pattern.

7.12.5.3. Pattern Procedures:

7.12.5.3.1. Include the type of practice emergency pattern in the gear check call.

7.12.5.3.2. Initiate practice single-engine go-arounds in sufficient time to ensure the aircraft does not descend below 300 feet AGL. Simulated single-engine approaches may descend below 300 feet AGL provided the approach terminates in a full stop landing or the go-around from a low approach or touch-and-go landing is performed with both engines. When conducting such training, the pilot will advise the appropriate air traffic controller of her or his intentions. **(T-2)**.

7.13. Search and Rescue Combat Air Patrol (SARCAP) Procedures. If an aircraft crashes, immediately attempt to locate possible survivors and initiate rescue efforts. Expect that the aircrews may initially suffer from shock or have delayed reactions due to ejection injuries. The following procedures are by no means complete and should be adjusted to meet each unique search and rescue situation.

7.13.1. Knock off maneuvering.

7.13.2. Establish a SARCAP commander.

7.13.3. Squawk 7700 to alert ATC/ground controlled intercept (GCI) of the emergency situation.

7.13.4. Communicate the emergency situation and aircraft/flight intentions immediately to applicable control agencies. Use GUARD frequency if necessary.

7.13.5. Mark the last known position of survivors/crash site using any means available (TACAN, EGI, FDL, ATC/GCI position and/or visual references).

7.13.6. Remain above the highest ejection altitude, if known, or the highest observed parachute until determining the position of all possible survivors.

7.13.7. Deconflict other aircraft assisting in the SARCAP by altitude to preclude midair collision. Establish high/low combat air patrol (CAP) as necessary to facilitate communications.

7.13.8. Revise bingo fuels or recovery bases as required to maintain maximum SARCAP coverage. Do not overfly the adjusted bingo fuel.

7.13.9. Relinquish SARCAP operation to designated rescue forces upon their arrival.

7.13.10. Follow local or briefed procedures.

7.14. Lateral Asymmetry.

7.14.1. When a fuel imbalance develops that exceeds 5000 ft./lbs. of lateral asymmetry, terminate maneuvering and investigate. If the imbalance was caused by a slow feeding external or internal wing tank, restrict maneuvering to instruments, deployment missions, or limited maneuvering intercepts less than 30 units AOA, in accordance with tech order guidance, until the imbalance is corrected. Pilots will use the high AOA warning tone as a tool to limit the aircraft to less than 30 units. **(T-2)**. If the imbalance cannot be corrected, terminate the mission. The above events can be flown to reduce gross weight.

7.14.2. Weapon and stores asymmetry must also be included to compute the total asymmetry condition. **(T-2)**.

7.14.3. Use the following rules of thumb in **Table 7.1** to determine if a fuel asymmetry condition exceeds the Dash-1 limit of 5000 ft/lbs:

Table 7.1. Asymmetry ROTs.

Location of Fuel	To get total ft/pounds:
Internal Wing Fuel	8 X imbalance = (ft/pounds)
External Wing Fuel	10 X imbalance = (ft/pounds)

Chapter 8

LOCAL OPERATING PROCEDURES

8.1. This chapter is reserved for unit local operating procedures.

8.1.1. Procedures herein will not be less restrictive than those contained elsewhere in this instruction. **(T-2)**. This chapter is not intended to be a single source document for procedures contained in other directives or regulations. Avoid unnecessary repetition of guidance provided in other established directives; however, reference to those directives is acceptable when it serves to facilitate location of information necessary for local operating procedures. Each pilot is authorized a copy of this chapter. MAJCOMs or other subordinate agencies (Numbered Air Force (NAF), Center, etc.), may direct publications approval channels and a specific format for **Chapter 8** based on unique flying areas, missions, and/or procedures. Unless changed by MAJCOM or subordinate agency, the following procedures apply:

8.1.1.1. When published, units will forward copies to MAJCOM and appropriate subordinate agencies, who will review **Chapter 8** and return comments or required changes back to the unit(s), if appropriate. **(T-2)**. The process need not delay distribution unless specified otherwise by MAJCOM or a subordinate agency. If a procedure is applicable to all F-15 A-D units, it will be incorporated into the basic instruction. **(T-2)**.

8.1.1.1.1. Organize the local chapter in the following format and, as a minimum, include the following:

8.1.1.1.1.1. Section A. Introduction.

8.1.1.1.1.2. Section B. General Policy.

8.1.1.1.1.3. Section C. Ground Operations.

8.1.1.1.1.4. Section D. Flying Operations.

8.1.1.1.1.5. Section E. Weapons Employment.

8.1.1.1.1.6. Section F. Abnormal Procedures.

8.1.1.1.1.7. Attachments. (Illustrations)

8.1.1.1.2. Include procedures for the following, if applicable:

8.1.1.1.2.1. Command and Control.

8.1.1.1.2.2. Fuel Requirements and Bingo Fuels.

8.1.1.1.2.3. Diversion Instructions.

8.1.1.1.2.4. Jettison Areas/Procedures/Parameters (IFR/VFR).

8.1.1.1.2.5. Controlled Bailout Areas.

8.1.1.1.2.6. Local Weather Procedures.

8.1.1.1.2.7. Unit Standards (Optional).

8.1.1.1.2.8. Approved Alternate Missions.

8.1.1.1.2.9. Cross-Country Procedures (if applicable).

8.1.1.1.2.10. Search and Rescue Procedures (SARCAP).

8.1.1.1.2.11. Local Environmental Restrictions to Flight Operations (winds, sea state, temperature, etc.) as applicable to unit locations.

8.1.2. Avoid unnecessary repetition of guidance provided in other established directives; however, reference to those directives is acceptable when it serves to facilitate location of information necessary for local operating procedures. Each pilot is authorized a copy of this chapter.

8.1.3. MAJCOMs or other subordinate agencies (Numbered Air Force, Center, etc.), may direct publications approval channels and a specific format for **Chapter 8** based on unique flying areas, missions, and/or procedures.

8.2. Procedures. Unless changed by MAJCOM or subordinate agency, the following procedures apply.

8.2.1. When published, units will forward copies to MAJCOM and appropriate subordinate agencies, who will review the **Chapter 8** and return comments or required changes back to the unit(s), if appropriate. **(T-2)**. The process need not delay distribution unless specified otherwise by MAJCOM or a subordinate agency. If a procedure is applicable to all F-15 A-D units, it will be incorporated into the basic instruction. **(T-2)**.

8.2.2. Organize the local chapter in the following format and, as a minimum, include the following:

8.2.2.1. Section A. Introduction.

8.2.2.2. Section B. General Policy.

8.2.2.3. Section C. Ground Operations.

8.2.2.4. Section D. Flying Operations.

8.2.2.5. Section E. Weapons Employment.

8.2.2.6. Section F. Abnormal Procedures.

8.2.2.7. Attachments. (Illustrations)

8.2.2.7.1. Include procedures for the following, if applicable:

8.2.2.7.2. Command and control.

8.2.2.7.3. Fuel Requirements and Bingo Fuels.

8.2.2.7.4. Diversion Instructions.

8.2.2.7.5. Jettison Areas/Procedures/Parameters (IFR/VFR).

8.2.2.7.6. Controlled Bailout Areas.

8.2.2.7.7. Local Weather Procedures. 8.1.1.1.2.7. Unit Standards (Optional).

8.2.2.7.8. Approved Alternate Missions.

8.2.2.7.9. Cross-Country Procedures (if applicable).

8.2.2.7.10. Search and Rescue Procedures (SARCAP).

8.2.2.7.11. Local Environmental Restrictions to Flight Operations (winds, sea state, temperature, etc.) as applicable to unit locations.

JOSEPH T. GUASTELLA Jr., Lt Gen, USAF
Deputy Chief of Staff, Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFMAN 11-202V3, *Flight Operations*, 10 June 2020

AFPAM 11-205, *Aircrew Quick Reference to Aircraft Cockpit and Formation Flight Signals,* 9 August 2018

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Prescribed Forms

None

Adopted Forms

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AF Form 847, *Recommendation for Change of Publication*

AFTO Form 781A, *Maintenance Discrepancy and Work Document*

DD Form 175, *Military Flight Plan*

DD Form 1801, *DoD International Flight Plan*

Abbreviations and Acronyms

AAI—Air-to-Air Interrogator

A/A—Air to Air

AATC—ANG AFRC Test Center

AB—After Burner

ACBT—Air Combat Training

ACC—Air Combat Command

ACCSUP—Air Combat Command Supplement

ACDE—Aircrew Chemical Defense Equipment

ACMI—Air Combat Maneuvering Instrumentation

AFAFRICA—Air Forces AFRICA

AFI—Air Force Instruction

AFFSA—Air Force Flight Standards Agency

AFMAN—Air Force Manual

AFPD—Air Force Policy Document

AFTO—Air Force Technical Order

AFTTP—Air Force Tactics, Techniques, and Procedures

A/G—Air to Ground

AGTS—Aerial Gunnery Target Set

AHC—Air Handling Characteristics

ANG—Air National Guard

AOA—Angle of Attack

AP/1B—Area Planning Military Training Route Charts

ARCP—Air Refueling Contact Point

ARCT—Air Refueling Control Time

ARIP—Air Refueling In Initial Point

ASR—Airport Surveillance Radar

ASRR—Airfield Suitability Report/Summary Airfield Restriction

ATC—Air Traffic Control

AWACS—Airborne Warning and Control System

BDA—Battle Damage Assessment

BSA—Basic Surface Attack

BVR—Beyond Visual Range

CAP—Combat Air Patrol

CBRNE—Chemical, Biological, Radiological, Nuclear, and High Yield Explosive

CC—Commander

CG—Center of Gravity

CMR—Combat Mission Ready

COMAFFOR—Commander, Air Force Forces

CONUS—Continental United States

CPT—Cockpit Procedures Trainer

C/S—Call Sign

CW—Chemical Warfare

C2—Command and Control

(D)—Dissimilar

DAFI—Department of the Air Force Instruction

DEEC—Digital Electronic Engine Control

DoD—Department of Defense

DVR—Digital Video Recorder

ECM—Electronic Countermeasures

EGI—Embedded Global Positioning System/Inertial Navigation System

EID—Electronic Identification

EMCON—Emissions Control

EOR—End of Runway

EO TDA—Electro Optical Target Density Altitude

EP—Emergency Procedures

FAM—Familiarization

FCIF—Flight Crew Information File

FDL—Fighter Data Link

FEBA—Forward Edge of Battle Area

FLIP—Flight Information Publications

FOD—Foreign Object Damage

FTIT—Fan Turban Inlet Temperature

FTD—Flying Training Deployment

FTU—Formal Training Unit

G—Gravitational Load Factor

GCE—Ground Crew Ensemble

GCI—Ground Controlled Intercept

GNSS—Global Navigation Satellite System

HAS—Hardened Aircraft Shelter (Chapter 2 only)

HAS—High Angle Strafe (Chapter 6 and Attachment 13)

HHQ—Higher Headquarters

HI—High-Illumination

HQ—Headquarters

HUD—Heads Up Display

ICAO—International Civil Aviation Organization

ID—Identification

IFE—Inflight Emergency

IFF—Identification Friend or Foe

IFR—Instrument Flight Rules

IMC—Instrument Meteorological Conditions

IP—Instructor Pilot

IPE—Individual Protective Equipment

IQT—Initial Qualification Training

JFS—Jet Fuel Starter

JOAP—Joint Oil Analysis Program

JHMCS—Joint Helmet Mounted Cueing System

KIAS—Knots Indicated Airspeed

KIO—"knock—it-off"

KM—Kilometers

KTAS—Knots True Airspeed

LEP—Laser Eye Protection

LOWAT—Low Altitude Training

MAJCOM—Major Command

MAS—Maritime Air Support

MDS—Mission Design Series

MIL/AB—Military/Afterburner

MIT—Mass Item

MOPP—Mission Oriented Protective Posture

MQT—Mission Qualification Training

MSA—Minimum Safe Altitude

MSL—Mean Sea Level

MTR—Military Training Route

N/A—Not Applicable

NAS—National Airspace System

NLT—No Later Than

NM—Nautical Miles

NORAD—North American Aerospace Defense Command

NORDO—No Radio

NOTAMS—Notices to Airmen

NVGs—Night Vision Goggles

OCONUS—Outside the Continental U.S.

OG—Operations Group

OPR—Office of Primary Responsibility

Ops—Operations

OSC—On-scene Commander

OT&E—Operational Test and Evaluation

OWS—Overload Warning System

PACAF—Pacific Air Forces

PAR—Precision Approach Radar

PDM—Program Depot Maintenance

PIC—Pilot in Command

PWC—Pilot Weather Category

RCP—Rear Cockpit

RCR—Runway Condition Reading/Report

RNAV—Area Navigation

ROE—Rules of Engagement

ROT—Rules of Thumb

RTB—Return to Base

RVSM—reduced vertical separation minimum

RWR—Radar Warning Receiver

SA—Situational Awareness

SAR—Search and Rescue

SARCAP—Search and Rescue Combat Air Patrol

SD—Spatial Disorientation

SEFE—Stan/Eval Flight Examiners

SETOS—Single Engine Takeoff Speed

SIF—Selective Identification Feature

SII—Special Interest Item

SIT—Situation Display

SPINS—Special Instructions

SOF—Supervisor of Flying

TACAN—Tactical Air Navigation

TD—Type Designator

TDU—Target Designation Unit

TDY—Temporary Duty

TERPS—Terminal Instrument Procedures

TOLD—Takeoff and Landing Data

T.O.—Technical Order

TRs—Training Rules

UK—United Kingdom

US—United States

US&P—United States and its possessions

USAF—United States Air Force

USAFE/AFAFRICA—United States Air Forces in Europe/Air Forces Europe

VA—Visual Augmenter

VID—Visual Identification

VFR—Visual Flight Rules

VMC—Visual Meteorological Conditions

VR/IR—Visual Route/ Instrument Route

VSD—Vertical Situation Display

WSEP—Weapon Systems Evaluation Program

Attachment 2

GENERAL BRIEFING GUIDE

A2.1. MISSION DATA.

A2.1.1. Time Hack

A2.1.2. EP/Threat of the Day

A2.1.3. Mission Objective(s)

A2.1.4. Mission Overview

A2.1.5. Mission Data Card

A2.1.5.1. Mission Commander/Deputy Lead

A2.1.5.2. Joker/Bingo Fuel

A2.1.5.3. Takeoff and Landing Data

A2.1.5.4. Working Area

A2.1.6. Weather/Sunrise/Sunset/Moon Illumination

A2.1.7. NOTAMs/Bird Strike Potential

A2.1.8. Personal Equipment

A2.1.9. FCIF/Pubs/Maps

A2.2. GROUND PROCEDURES.

A2.2.1. Pre-Flight

A2.2.1.1. Aircraft

A2.2.1.2. Armament

A2.2.2. Check-In

A2.2.3. Taxi/Marshaling/Arming

A2.2.4. Spare Procedures

A2.3. TAKEOFF.

A2.3.1. Runway Lineup

A2.3.2. Takeoff Interval

A2.3.3. Abort

A2.3.4. Jettison Procedures

A2.3.5. Low Altitude Ejection

A2.3.6. Landing Immediately After Takeoff

A2.4. DEPARTURE/ENROUTE.

A2.4.1. Routing

A2.4.2. Trail Departure

A2.4.3. Join-Up/Formation

A2.4.4. Systems/Ops Checks

A2.5. RECOVERY.

A2.5.1. Rejoin

A2.5.2. Battle Damage Check

A2.5.3. Type Recovery

A2.5.4. Flight Break-Up

A2.5.5. Pattern and Landing A2.5.6. After Landing/De-Arm

A2.5.6. Emergency/Alternate Airfields

Attachment 3

SPECIAL SUBJECT BRIEFING GUIDE

A3.1. Instructor Responsibilities.

A3.2. Chase Procedures

A3.3. IFF Procedures

A3.4. Radar/Visual Search Responsibilities/Midair Collision Avoidance

A3.5. Dissimilar Formations

A3.6. Carriage/Jettison Limitations

A3.6.1. External Stores Limitations

A3.6.2. Bay 5 Stowage Requirements/Limitations

A3.7. Terrain Avoidance

A3.7.1. Departure/Enroute/Recovery

A3.7.2. Use of Radar Altimeters/MSL Floor Settings

A3.8. Bird Strike Procedures/Use of Visor(s)

A3.9. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/ Prioritization, and Complacency)

A3.10. G-Awareness

A3.10.1. G-Suit connection/G-tolerance/G-Awareness Turn

A3.10.2. Use of L-1 Anti-G Straining Maneuver

A3.11. Visual Illusions/Perceptions

A3.12. Spatial Disorientation/Unusual Attitudes

A3.13. Lost Wingman

A3.14. Radio Inoperative

A3.15. SARCAP

A3.16. Recall Procedures

A3.17. SIIs

Attachment 4

ADVANCED HANDLING/INSTRUMENT BRIEFING GUIDE

A4.1. AIRWORK

A4.1.1. Airspace Restrictions

A4.1.2. Area Orientation

A4.1.3. Instructor Responsibilities

A4.1.4. Maneuvers

A4.2. APPROACHES

A4.2.1. Frequencies

A4.2.2. Holding

A4.2.3. Penetration

A4.2.4. Missed Approach/Climb Out

A4.3. SPECIAL SUBJECTS

A4.3.1. "G" Awareness

A4.3.2. Fuel Awareness/AB Use/Consumption Rates

A4.3.3. Maneuvering Limitations

A4.3.3.1. Airspeed and "G"

A4.3.3.2. Recognition/Prevention/Recovery From Out of Control

A4.3.3.3. Maneuvering at Heavyweight/High Angles of Attack/Asymmetrical Configuration

A4.3.3.4. Effects of CG Throughout the Flight

A4.3.3.5. Time to Ground Impact

A4.3.3.5.1. Wings Level

A4.3.3.5.2. Overbank/Under G

A4.3.4. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 5

AIR REFUELING BRIEFING GUIDE

A5.1. GENERAL

A5.1.1. Tanker Call Sign(s)/Receiver Assignments

A5.1.2. Refueling Track(s)

A5.1.2.1. Altitude

A5.1.2.2. Airspeed

A5.1.2.3. Airspace Restrictions

A5.1.3. ARIPs, ARCPs, ARCTs

A5.1.4. Radio Frequencies

A5.2. BUDDY PROCEDURES

A5.2.1. Departure

A5.2.2. Join-Up

A5.3. ENROUTE

A5.3.1. Route of Flight

A5.3.2. Formation

A5.3.3. Ops Checks

A5.4. RENDEZVOUS

A5.4.1. Type Rendezvous

A5.4.2. Holding Procedures/Formation

A5.4.3. Ground Radar Assistance

A5.4.4. Tanker Identification - TACAN/Radar/AAI/FDL/Visual

A5.4.5. Radar Procedures/Techniques

A5.4.6. Wingman/Deputy Lead Responsibilities

A5.4.7. Receiver Formation/Join-Up Procedures

A5.4.8. Rendezvous Overrun

A5.5. REFUELING

A5.5.1. Checklist Procedures

A5.5.2. Radio Calls

A5.5.3. Refueling Order

A5.5.4. Techniques

A5.5.5. Radio Silent Procedures

A5.5.5.1. EMCON

A5.5.5.2. Visual Signals

A5.5.6. Fuel Off-Load

A5.5.7. Bingo Fuel (Abort Points/Abort Bases)

A5.5.8. Drop-Off Procedures

A5.5.9. Wake Turbulence

A5.6. REFORM AND EXIT

A5.6.1. Formation

A5.6.2. Clearance

A5.7. EMERGENCY PROCEDURES

A5.7.1. Breakaway Procedures

A5.7.2. Systems Malfunctions

A5.7.3. Damaged Receptacle

A5.8. IMC/NIGHT CONSIDERATIONS

A5.8.1. Lost Wingman Procedures

A5.8.2. Aircraft Lighting

A5.9. SPECIAL SUBJECTS

A5.9.1. Fuel Awareness/AB Use/Consumption Rates

A5.9.2. Flight Path Deconfliction/Other Receiver Considerations

A5.9.3. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 6

AIR COMBAT TRAINING (ACBT) / INTERCEPT BRIEFING GUIDE

A6.1. GENERAL/ADVERSARY COORDINATION/GCI COORDINATION

A6.1.1. Call Signs

A6.1.2. Number and Type Aircraft

A6.1.3. Scenario

A6.1.3.1. Objective(s)

A6.1.3.2. Type Threat Simulated/Tactics Limitations (If Any)

A6.1.3.3. CAP Points/Target Locations

A6.1.3.4. Safe Areas/FEBA/Ground Threats

A6.1.3.5. VID/EID/BVR Criteria

A6.1.4. Mission Contingencies

A6.1.4.1. Single Radar Scope/No GCI

A6.1.4.2. Single Frequency

A6.1.4.3. Area Weather/Alternate Mission

A6.1.4.4. Minimum Participants (Primary/Alternate Missions)

A6.1.5. Area Information

A6.1.5.1. Controlling Agency (GCI/AWACS/ACMI)

A6.1.5.1.1. GCI/Flight

A6.1.5.1.2. Comm Requirements

A6.1.5.1.3. Type/Level of Control

A6.1.5.2. Airspace Restrictions

A6.1.5.3. CAP Points/Target Locations

A6.1.5.4. Frequencies

A6.1.5.5. Squawks

A6.1.5.6. Block Altitudes/Min Altitudes/Flight Parameters

A6.1.5.7. Transmissions

A6.1.5.7.1. KIO

A6.1.5.7.2. Shots/Kills

A6.1.5.7.3. Fuel/Altitude Awareness

A6.1.6. Rendezvous/Recovery Procedures

A6.1.6.1. Dissimilar Formation

A6.1.7. Weapons Employment

A6.1.7.1. Simulated Ordnance (Type/Quantity)

A6.1.7.2. Shot Criteria

A6.1.7.3. Kill Criteria/Removal

A6.1.7.4. Shot/Kill Passage

A6.1.8. Training Rules

A6.1.9. Emergency Procedures

A6.1.9.1. Recovery

A6.1.9.2. Escort Procedures

A6.1.10. Debriefing (Time/Place)

A6.2. FLIGHT/ELEMENT TACTICS

A6.2.1. Avionics Set-up

A6.2.1.1. Radar

A6.2.1.2. EGI

A6.2.1.3. IFF

A6.2.1.4. Air-to-air TACAN

A6.2.1.5. FDL

A6.2.2. CAP/Patrol Phase

A6.2.2.1. Type Pattern

A6.2.2.2. Formation/Altitude/Airspeed

A6.2.2.3. Search Responsibilities

A6.2.2.4. Commit

A6.2.2.4.1. Criteria/Range

A6.2.2.4.2. Procedures

A6.2.3. Ingress/Intercept Phase

A6.2.3.1. Formation/Altitude/Airspeed

A6.2.3.2. Detection

A6.2.3.2.1. Search Responsibilities (Radar/Visual)

A6.2.3.2.2. Radar Sorting

A6.2.3.3. Targeting Plan

A6.2.3.4. Intercept Type/Planned Tactics

A6.2.3.4.1. Plan (Direct Attack/Deception)

A6.2.3.4.2. Mutual Support Requirements

A6.2.3.4.3. Identification Requirements/Procedures

A6.2.3.4.4. Minimum Altitudes/Airspeeds

A6.2.3.4.5. Vertical/Horizontal Conversions/Turning Room

A6.2.3.5. Night/IMC Intercepts

A6.2.3.5.1. ECM/Chaff/ Evasion Restrictions

A6.2.3.5.2. Radar Requirements

A6.2.3.5.3. Altitude Separation Requirements

A6.2.4. Engagement Phase

A6.2.4.1. Plan

A6.2.4.1.1. Turn and Fight

A6.2.4.1.2. Hit and Run

A6.2.4.1.3. Abort

A6.2.4.2. Clearance for Wingman to Engage

A6.2.4.2.1. Offensive

A6.2.4.2.2. Defensive

A6.2.4.3. Alternate Plan (Degraded Situation)

A6.2.5. Egress/Separation Phase

A6.2.5.1. Disengagement Plan (Why/When/How)

A6.2.5.1.1. Loss of Mutual Support

A6.2.5.1.2. Fuel

A6.2.5.1.3. Ordnance

A6.2.5.2. Egress Formation/Responsibilities

A6.2.6. Contingencies

A6.2.6.1. Single Contact

A6.2.6.2. Short Range Commit

A6.2.6.3. Single Ship (Loss of Mutual Support)

A6.2.6.4. Safe Escape/Rendezvous Point

A6.2.7. Live Missile/Hot Gun Safety Procedures

A6.2.8. Additional Considerations

A6.2.8.1. Threat Reaction

A6.2.8.2. Degraded Systems

A6.2.8.3. Tactical Lead Changes

A6.2.8.4. Bandit Options

A6.2.8.5. Film/DVR

A6.2.8.6. Codewords

A6.2.8.7. Environmental Considerations

A6.2.9. Alternate Mission

A6.2.9.1. Type Mission (refer to appropriate mission briefing guide)

A6.2.9.2. Mission Objectives

A6.3. SPECIAL SUBJECTS

A6.3.1. "G" Awareness

A6.3.2. Fuel Awareness/AB Use/Consumption Rates

A6.3.3. Flight Path Deconfliction

A6.3.4. Maneuvering Limitations

A6.3.4.1. Airspeed and "G"

A6.3.4.2. Recognition/Prevention/Recovery from Out of Control

A6.3.4.3. Time to Ground Impact

A6.3.4.3.1. Wings Level

A6.3.4.3.2. Overbank/Under "G"

A6.3.5. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 7

ESCORT MISSION BRIEFING GUIDE

A7.1. ENROUTE TO RENDEZVOUS/POST-MISSION NAVIGATION

A7.1.1. Formation

A7.1.2. Route of Flight

A7.1.3. Control Agency Call Sign/Frequency

A7.2. RENDEZVOUS

A7.2.1. Protected Force Call Sign

A7.2.2. Altitude

A7.2.3. Airspeed

A7.3. ESCORT PROCEDURES

A7.3.1. Type Formation

A7.3.2. Tactics

A7.3.3. Commit

A7.3.3.1. Criteria/Range

A7.3.3.2. Procedures

A7.3.4. Escort Route

A7.3.5. ECM/RWR

A7.4. TRAINING RULES

A7.5. ALTERNATE MISSION

A7.5.1. Type Mission (refer to appropriate mission briefing guide)

A7.5.2. Mission Objectives

A7.6. SPECIAL SUBJECTS

A7.6.1. Airspace Restrictions

A7.6.2. “G” Awareness

A7.6.3. Fuel Awareness/AB Use/Consumption Rate

A7.6.4. Flight Path Deconfliction

A7.6.5. Maneuvering Limitations

A7.6.5.1. Airspeed and “G”

A7.6.5.2. Recognition/Prevention/Recovery from Out of Control

A7.6.6. Time to Ground Impact

A7.6.6.1. Wing Level

A7.6.6.2. Overbank/Under “G”

A7.6.7. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 8

AERIAL GUNNERY TOW COORDINATION BRIEFING GUIDE

A8.1. TOW COORDINATION

A8.1.1. Ground/Takeoff./Departure

A8.1.2. Rendezvous

A8.1.3. Airspace Data

A8.1.4. GCI Support

A8.1.5. Target Launch/Chase

A8.1.6. Shooter Order

A8.1.7. Type Pattern

A8.1.8. Tow Altitude Block(s)/Flight Parameters

A8.1.9. Intercept Phase/Pattern Set-Up

A8.1.10. Arming Procedures

A8.1.11. Timing

A8.1.12. Tow Maneuvering Parameters

A8.1.13. Shooter/Firing Plan

A8.1.14. Radio Procedures

A8.1.15. Termination

A8.1.15.1. Timing

A8.1.15.2. Minimum Altitude

A8.1.15.3. Joker/Bingo Fuel

A8.1.15.4. Winchester

A8.1.15.5. Fouls

A8.1.16. Armament Safety Check

A8.1.17. Scoring

A8.1.18. Subsequent Set-Ups

A8.1.19. Target Drop Procedures

A8.1.20. Recovery Order

A8.1.21. Abnormal Procedures

A8.1.21.1. Erratic Target

A8.1.21.1.1. During Deployment

A8.1.21.1.2. During Employment

A8.1.21.2. Target Drag-Off

A8.1.21.3. Recovery With Target/Cable

A8.1.21.4. NORDO

A8.1.21.4.1. During Engagement

A8.1.21.4.2. Target Drop

A8.1.21.4.3. Visual Signals

A8.1.21.4.4. Recovery

Attachment 9

AERIAL GUNNERY BRIEFING GUIDE

A9.1. GENERAL

A9.1.1. Formation

A9.1.2. Area Information

A9.1.2.1. Controlling Agency

A9.1.2.2. Airspace Restrictions

A9.1.2.3. Frequencies

A9.1.3. Switch Positions

A9.1.4. Arming Procedures

A9.1.5. Intercept/Set-Up

A9.1.6. Shooter Sequence

A9.1.7. Position Changes

A9.1.8. Chase Procedures

A9.1.9. Timing

A9.2. EMPLOYMENT

A9.2.1. Firing Parameters

A9.2.1.1. Minimum Range

A9.2.1.2. Overtake

A9.2.1.3. Angle-Off

A9.2.1.4. Error Analysis

A9.2.2. Contingencies

A9.2.2.1. Avionics Malfunctions

A9.2.2.2. Gun Malfunctions

A9.2.2.3. Range Estimation Without Radar

A9.2.3. Safety Considerations

A9.2.3.1. Target Fixation

A9.2.3.2. Debris Avoidance

A9.2.3.3. Fouls

A9.3. TRAINING RULES

A9.4. SPECIAL INTEREST ITEMS

A9.5. ALTERNATE MISSION

A9.5.1. Type Mission (refer to appropriate mission briefing guide)

A9.5.2. Mission Objectives

A9.6. SPECIAL SUBJECTS

A9.6.1. Minimum Altitudes

A9.6.2. “G” Awareness

A9.6.3. Fuel Awareness/Ops Checks/AB Use/Consumption Rates

A9.6.4. Maneuvering Limitations

A9.6.4.1. Airspeed/”G”/Stress

A9.6.4.2. Recognition/Prevention/Recovery From Out of Control

A9.6.5. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 10

LOW-LEVEL NAVIGATION BRIEFING GUIDE

A10.1. GENERAL

A10.1.1. Route/Clearance/Restrictions

A10.1.2. Flight Responsibilities

A10.1.2.1. Navigation

A10.1.2.2. Radar/Visual Search

A10.1.3. Entry/Spacing /Holding/Initial Altitude (MSA)

A10.2. ROUTE PROCEDURES

A10.2.1. Fence Checks

A10.2.2. Tactical Formation/Turns

A10.2.3. Low Level Navigation

A10.2.3.1. Dead Reckoning/Use of Navigation Aids/ Equipment (i.e., EGI/SIT)

A10.2.3.2. Radar Procedures/Techniques

A10.2.3.3. Visual Procedures/Techniques

A10.2.3.4. Updates/Calibrations

A10.2.3.5. Time/Fuel Control

A10.2.3.6. Terrain Following/Avoidance/Wingman Considerations

A10.2.3.7. Leg Altitudes/Set Clearance Plane/Obstacles (MSL/AGL)

A10.2.3.8. Turnpoint Acquisition

A10.2.4. Threat Reactions

A10.2.4.1. RWR/ECM/Chaff/Flares

A10.2.4.2. Engagement Criteria

A10.2.4.3. Flight Path Deconfliction

A10.2.4.4. Termination

A10.3. EMERGENCIES

A10.3.1. Aircraft Malfunctions

A10.3.2. Route Abort Procedures (ERAA/MSA)/ATC Frequencies

A10.4. TRAINING RULES

A10.5. SPECIAL INTEREST ITEMS

A10.6. ALTERNATE MISSION

A10.6.1. Type mission (refer to appropriate mission briefing guide)

A10.6.2. Mission Objectives

A10.7. SPECIAL SUBJECTS

A10.7.1. Airspace Restrictions

A10.7.2. "G" Awareness/Ops Checks

A10.7.3. Fuel Awareness/AB Use/Consumption Rates

A10.7.4. Flight Path Deconfliction

A10.7.5. Maneuvering Limitations

A10.7.5.1. Airspeed and "G"

A10.7.5.2. Recognition/Prevention/Recovery From Out of Control

A10.7.6. Time to Ground Impact

A10.7.6.1. Wings Level

A10.7.6.2. Overbank/Under "G"

A10.7.7. Night Considerations

A10.7.8. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 11

NVGS BRIEFING GUIDE

A11.1. This guide is meant to highlight general NVGs considerations, and provides a reference for a basic NVGs briefing. All applicable NVGs considerations should be incorporated into the specific briefing for the mission being flown.

A11.2. WEATHER/ILLUMINATION

A11.2.1. Civil/Nautical Twilight

A11.2.2. Moon Rise/Set Times/Phase/Elevation/Azimuth

A11.2.3. Ceiling/Visibility

A11.2.4. Lux/EO TDA

A11.2.5. Obscurants to Visibility

A11.3. NVGs PREFLIGHT

A11.3.1. Check Adjustments/Helmet Fit and Security

A11.3.2. Batteries

A11.3.3. Resolution/Focus

A11.3.4. NVGs-Compatible Flashlight

A11.4. COCKPIT PREFLIGHT

A11.4.1. Cockpit Setup

A11.4.2. Cockpit Lighting (Leaks)/Mirrors Up

A11.4.3. Cockpit FAM

A11.4.4. Check Focus and Stow for Taxi

A11.5. BEFORE TAKEOFF

A11.5.1. Don NVGs/Check and Adjust/Disconnect

A11.5.2. Use During Taxi

A11.5.3. Stow for Takeoff

A11.6. AIRBORNE

A11.6.1. Exterior Lights

A11.6.2. Donning and Doffing Procedures

A11.6.3. Scan Pattern

A11.6.3.1. Forward Scan

A11.6.3.2. Narrow Field of View

A11.6.3.3. Peripheral Vision

A11.6.3.4. Scan Techniques

A11.6.4. Join-up and Enroute Considerations
A11.6.4.1. Rejoin/Closure
A11.6.4.2. Air to Air TACAN/FDL
A11.6.4.3. G-Awareness Considerations
A11.6.4.3.1. Lighting
A11.6.4.3.2. Deconfliction/Separation
A11.7. MISSION
A11.7.1. Route Study/Scene Interpretation
A11.7.1.1. NVGs Predictions/Albido
A11.7.1.2. Terrain/Shadowing/Visual Illusions/Visible Horizon
A11.7.1.3. City/Cultural Lighting
A11.7.1.3.1. Direction/Orientation of Lighting
A11.7.1.3.2. Aggressive Formation Maneuvering
A11.7.1.3.3. Terrain Avoidance
A11.7.1.3.4. Map Reading
A11.8. TARGET AREA
A11.8.1. RV/Holding Procedures (NVGs Differences)
A11.8.2. NVGs Lost Wingman
A11.8.3. Deliveries/Pattern Procedures
A11.8.3.1. Minimum Altitudes
A11.8.3.2. Flight Member Responsibilities
A11.8.3.3. Moth Effect
A11.8.3.4. Deconfliction
A11.8.3.5. External Lighting/Deconfliction Procedures
A11.8.4. Threat ID and Reaction
A11.8.5. Egress
A11.9. NVGs SAFETY
A11.9.1. NVGs Lost Sight
A11.9.2. NVGs Lost Wingman
A11.9.3. Depth Perception
A11.9.4. Visual Illusions
A11.9.5. NVGs Failure

A11.9.6. Battery Failure/Swap Out

A11.9.7. Overconfidence in NVGs Capabilities

A11.9.8. Entering Weather/Transition to Instruments

A11.9.9. Correct Lighting of Primary/Secondary Flight Instruments

A11.9.10. Disorientation/Misorientation/Vertigo

A11.9.11. Deconfliction ROE

A11.9.12. Transference

A11.9.13. Target Fixation

A11.9.14. Lack of Attitude Information

A11.9.15. Fatigue

A11.9.16. Aircraft Emergency and NVGs BDA Considerations

A11.9.17. Ejection/Goggles Off

A11.9.18. NVGs Foreign object damage (FOD) Considerations (Batteries, Equipment, Filters, etc.)

A11.9.19. Laser Eye Protection (LEP) use

Attachment 12

ALERT BRIEFING GUIDE

A12.1. This guide is all-inclusive and is designed to incorporate all the applicable items from the General Briefing Guide. If a specialized mission such as air refueling is anticipated, the specific briefing guide for that mission should also be used.

A12.2. MISSION DATA

A12.2.1. Time Hack

A12.2.2. Mission Data Card

A12.2.2.1. Call Signs

A12.2.2.2. Aircraft/Location/Status

A12.2.2.3. Takeoff/Landing Data (Worst Case)

A12.2.2.4. Joker/Bingo Fuel

A12.2.3. Actual/Forecast Weather

A12.2.3.1. Home Base

A12.2.3.2. Alternates

A12.2.3.3. Individual Weather Category/Mandatory Status

A12.2.4. NOTAMs

A12.2.5. FCIF/Pubs/Maps

A12.2.6. Personal Equipment

A12.2.7. Alert Packet

A12.2.7.1. Authenticators/Duress Code

A12.2.7.2. Security Procedures

A12.2.8. Airfield Status

A12.2.8.1. Actual versus Max Allowable Tailwind

A12.2.8.2. Barriers

A12.2.8.3. Navigation Aids

A12.2.8.4. Hazards to Taxi/RCR

A12.3. GROUND PROCEDURES

A12.3.1. Aircraft/Armament Preflight

A12.3.2. Cockpit Set-Up

A12.3.3. Engine Run/Hot Preflight

A12.3.4. Crew Chief Briefing

A12.3.5. Quick Check Procedures

A12.4. LAUNCH PROCEDURES

A12.4.1. Notification/UHF Frequency/Authentication Requirement

A12.4.2. Status

A12.4.2.1. Airborne Order

A12.4.2.2. Battle Stations

A12.4.2.3. Runway Alert

A12.4.2.4. Scramble

A12.4.3. Taxi

A12.4.4. Takeoff/Runway Lineup/Interval/Formation

A12.4.4.1. Day VMC/IMC

A12.4.4.2. Night VMC/IMC

A12.4.5. Join-up/Trail Formation/Power Settings/Airspeeds

A12.5. IN-FLIGHT PROCEDURES

A12.5.1. Formation

A12.5.2. Airspeeds

A12.5.3. Weapons Safe Checks

A12.5.4. Radar Search Responsibilities

A12.5.5. Degraded Fire Control System

A12.5.6. Transfer of Lead Procedures

A12.5.7. Ops Checks

A12.5.8. EMCON Procedures

A12.5.9. Region Minimum Safe Altitude (MSA)

A12.5.10. VID Procedures

A12.5.10.1. Authority Required to Close

A12.5.10.2. Formation/Tactics

A12.5.10.3. Range/Altitude Separation Requirements on Target Prior Permission to Close With/Without Visual Contact

A12.5.10.4. Radar Lock-On Requirements

A12.5.10.5. Maximum Closure Speed

A12.5.10.6. Minimum Airspeed

A12.5.10.7. Loss of Contact Procedures

A12.5.10.8. Breakaway Procedures

A12.5.10.9. Restrictions

A12.5.11. Aircraft in Distress

A12.5.11.1. Minimum Closure Distance

A12.5.11.2. Visual Signals - Day/Night

A12.5.11.3. Escort Procedures

A12.5.11.4. Recovery/Landing Visual Signals

A12.5.11.5. Dissimilar Formation Procedures

A12.5.12. Jettison Procedures

A12.5.13. Lost Wingman

A12.5.14. SARCAP

A12.5.15. Emergency Airfields

A12.6. SPECIAL SUBJECTS

A12.6.1. Emergency of the Day

A12.6.2. Fuel Awareness

A12.6.3. Maneuvering Limitations

A12.6.4. Recognition/Prevention/Recovery from Loss of Control

A12.6.5. Spatial Disorientation

A12.6.6. Recall Procedures

A12.6.7. Rules of Engagement (ROE)

A12.6.8. Hazards Associated with Human Factors (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

A12.7. GROUND CREW BRIEFING

A12.7.1. Act only on pilot’s instructions

A12.7.2. Ground emergency procedures

A12.7.3. Hand signals

A12.7.4. Aircraft danger areas

Attachment 13

HIGH ANGLE STRAFE (HAS) BRIEFING GUIDE

A13.1. GENERAL

A13.1.1. Range/Off-Range Information

A13.1.2. Target/Range/Airspace Description

A13.1.3. Restrictions

A13.1.4. Range/Airspace Entry/Holding

A13.1.5. Radio Procedures

A13.1.6. Formation

A13.1.7. Sequence of Events

A13.1.8. Pattern Procedures

A13.1.9. Aircraft Fallout Plan

A13.1.10. Rejoin on Range for Late Takeoffs

A13.2. EMPLOYMENT

A13.2.1. Employment Procedures/Techniques: Avionics/Switch Positions

A13.2.2. Weapons Switchology

A13.2.3. Radar Switchology

A13.2.4. Ground track/Altitude/Airspeed

A13.2.5. Radar/Techniques

A13.2.6. Entry Airspeed/Altitude

A13.2.7. Target Acquisition

A13.2.8. Pattern Corrections

A13.2.9. Roll-In

A13.2.10. Position

A13.2.11. Techniques (Pitch/Bank/Power) Roll-Out/Wind Effect

A13.2.12. Final

A13.2.13. Dive Angle – 25 degrees for Strafe

A13.2.14. Airspeed

A13.2.15. HUD Depiction

A13.2.16. Sight Picture/Corrections/Aim-Point

A13.2.17. Trigger Actuation Parameters

A13.2.18. Climbing Safe Escape Maneuver Procedures

A13.2.19. Live Ordnance Considerations
A13.2.20. Safe Escape/Safe Separation/Frag Avoidance
A13.2.21. Flare Pattern
A13.2.22. Flare Release Points and Interval
A13.2.23. Wind Effect/Offset
A13.2.24. Dud Flare Procedures
A13.2.25. Switching Aircraft Patterns
A13.3. OVER WATER OPERATIONS
A13.3.1. Employment Techniques
A13.3.2. Depth Perception / Reduced Visual Cues
A13.3.3. Distance/Altitude Estimation
A13.3.4. Timing
A13.3.5. Visual/Aircraft References to Establish Pattern
A13.3.6. Special Considerations
A13.3.7. Adjusted Minimum Altitudes
A13.3.8. Moving targets
A13.4. RANGE DEPARTURE PROCEDURES
A13.4.1. Armament Safety Checks
A13.4.2. Rejoin
A13.5. ABNORMAL PROCEDURES
A13.5.1. Battle Damage
A13.5.2. Inadvertent Release
A13.5.3. Gun Unsafe/Jam/Hot Gun
A13.6. TRAINING RULES
A13.7. SPECIAL INTEREST ITEMS
A13.8. ALTERNATE MISSION
A13.9. SPECIAL SUBJECTS
A13.9.1. Error Analysis
A13.9.2. Fouls
A13.9.3. Minimum Altitudes
A13.9.4. Target Fixation
A13.9.5. G-Awareness

A13.9.6. Fuel Awareness/Ops Checks/AB Use/Consumption Rates

A13.9.7. Maneuvering Limitations

A13.9.8. Airspeed/G

A13.9.9. Recognition/Prevention/Recovery from Out of Control

A13.9.10. Time to Ground Impact

A13.9.11. Wings Level Overbank/Under G

A13.9.12. Human Factors Considerations (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 14

MARITIME AIR SUPPORT (MAS) AND BASIC SURFACE ATTACK (BSA) BRIEFING GUIDE

A14.1. GENERAL

A14.1.1. Range/Off-Range Procedures

A14.1.2. Target/Range/Airspace Description

A14.1.3. Restrictions

A14.1.4. Range/Airspace Entry/Holding

A14.1.5. Radio Procedures

A14.1.6. Formation

A14.1.7. Sequence of Events

A14.1.8. Aircraft Fallout Plan

A14.2. EMPLOYEMENT

A14.2.1. Employment Procedures/Techniques: Avionics/Switch Positions

A14.2.2. Weapons Switchology/Delivery Mode

A14.2.3. Radar Switchology

A14.2.4. Ground Track/Altitude/Airspeed

A14.2.5. Radar/Techniques

A14.2.6. Backup Deliveries

A14.2.7. Entry Airspeed/Altitude

A14.2.8. Target Acquisition

A14.2.9. Roll-In

A14.2.10. Position

A14.2.11. Techniques (Pitch/Bank/Power) Roll-Out/Wind Effect

A14.2.12. Final

A14.2.13. Shot Altitude/Strafe Dive Angle – 25 degrees

A14.2.14. Airspeed

A14.2.15. HUD Depiction

A14.2.16. Sight Picture/Corrections/Aim-Point

A14.2.17. Trigger Actuation

A14.2.18. Pickle Parameters

A14.2.19. Release Indications

A14.2.20. Climbing Safe Escape Maneuver Procedures

A14.2.21. Special Procedures:

A14.2.21.1. Live Ordnance Considerations Safe Escape/Safe Separation Fuse Arming/Frag Avoidance

A14.2.21.2. Flare Pattern

A14.2.21.3. Flare Release Points and Interval

A14.2.21.4. Wind Effect/Offset

A14.2.21.5. Dud Flare Procedures

A14.2.21.6. Switching Aircraft Patterns

A14.3. OVER WATER OPERATIONS

A14.3.1. Employment Techniques

A14.3.2. Depth Perception/Reduced Visual Cues

A14.3.3. Distance/Altitude Estimation

A14.3.4. Timing

A14.3.5. Visual/Aircraft References to Establish Pattern

A14.3.6. Special Considerations

A14.3.7. Adjusted Minimum Altitudes

A14.3.8. Moving Targets

A14.3.9. Range/Airspace Departure Procedures: Armament Safety Checks

A14.3.10. Rejoin

A14.4. ABNORMAL PROCEDURES

A14.4.1. Battle Damage/Parameters Hung/Unexpended Ordnance

A14.4.2. Inadvertent Release

A14.4.3. Gun Unsafe/Jam/Hot Gun

A14.5. TRAINING RULES

A14.6. SPECIAL INTEREST ITEMS

A14.7. ALTERNATE MISSION

A14.8. SPECIAL SUBJECTS

A14.8.1. Error Analysis

A14.8.2. Fouls

A14.8.3. Minimum Altitudes

A14.8.4. Target Fixation

A14.8.5. G-Awareness

A14.8.6. Fuel Awareness/Ops Checks/AB Use/Consumption Rates

A14.8.7. Maneuvering Limitations

A14.8.8. Airspeed/G/Stress (Carriage/Release)

A14.8.9. Recognition/Prevention/Recovery from Out of Control

A14.8.10. Time to Ground Impact Wings Level Overbank/Under G

A14.8.11. Human Factors Considerations (e.g., Channelized Attention, Task Saturation/Prioritization, and Complacency)

Attachment 15

CREW COORDINATION/PASSENGER/GROUND CREW BRIEFING GUIDE

A15.1. GROUND OPERATIONS

A15.1.1. Ramp Safety (danger areas/hearing and eye protection)

A15.1.2. Foreign Object Damage (FOD) Considerations

A15.1.3. Normal Ingress and Egress

A15.1.4. Strap-in Procedures/Proper Use of Restraints

A15.1.5. Life Support Equipment

A15.1.6. Oxygen System - Preflight and Normal Settings

A15.1.7. Canopy Procedures

A15.1.8. Ejection Seat Procedures

A15.1.9. Critical Switches and Controls

A15.1.10. Safety Precautions (e.g., stick/leg interference)

A15.1.11. Prohibitions and Restrictions

A15.1.12. Communications Connections and Use

A15.2. FLIGHT OVERVIEW AND PROFILE

A15.2.1. Takeoff and Departure

A15.2.2. Route, Air Work, Maneuvers

A15.2.3. Anti-G Straining Maneuver

A15.2.4. Transfer of Aircraft Control

A15.2.5. Recovery, Pattern, and Landing

A15.2.6. In-flight Checks (challenge and response)

A15.3. ABNORMAL PROCEDURES

A15.3.1. Emergency Ground Egress

A15.3.2. Abort

A15.3.3. In-flight Emergency Procedures

A15.3.4. Bird Strike

A15.3.5. Smoke and Fume Elimination

A15.3.6. Physiological

A15.3.7. Ejection/bail out

A15.3.8. Intercom Failure

A15.3.9. Oxygen Emergency Procedures - Confirm Passenger Demonstrates Mask Operation and Proper Regulator Settings

A15.4. GROUND CREW

A15.4.1. Ground Emergency Procedures

A15.4.2. Hand Signals

A15.4.3. Aircraft Danger Areas

Attachment 16

MISSION DEBRIEFING GUIDE

A16.1. GROUND PROCEDURES

A16.2. MISSION DEBRIEFING GUIDE

A16.3. TAKEOFF/JOIN-UP/DEPARTURE

A16.4. ENROUTE PROCEDURES

A16.5. RECOVERY/LANDING/AFTER LANDING

A16.6. GENERAL

A16.6.1. SIIs

A16.6.2. Radio Procedures

A16.6.3. Flight Discipline/Effectiveness

A16.7. MISSION ACCOMPLISHMENT/ANALYSIS

A16.7.1. Mission Reconstruction

A16.7.2. Mission Support

A16.7.3. DVR Assessment

A16.7.4. Anti-G Straining Maneuver Effectiveness

A16.7.5. Learning Objectives Achieved

A16.7.6. Lessons Learned

A16.7.7. Recommendations for Improvement

A16.8. COMMENTS/QUESTIONS